

Treatment for beta-blocker poisoning: a systematic review

Joe-Anthony Rotella, Shaun L. Greene, Zeff Koutsogiannis, Andis Graudins, Yit Hung Leang, Kelvin Kuan, Helen Baxter, Elyssia Bourke & Anselm Wong

To cite this article: Joe-Anthony Rotella, Shaun L. Greene, Zeff Koutsogiannis, Andis Graudins, Yit Hung Leang, Kelvin Kuan, Helen Baxter, Elyssia Bourke & Anselm Wong (2020): Treatment for beta-blocker poisoning: a systematic review, Clinical Toxicology, DOI: [10.1080/15563650.2020.1752918](https://doi.org/10.1080/15563650.2020.1752918)

To link to this article: <https://doi.org/10.1080/15563650.2020.1752918>



Published online: 20 Apr 2020.



Submit your article to this journal [↗](#)



Article views: 3



View related articles [↗](#)



View Crossmark data [↗](#)

REVIEW ARTICLE



Treatment for beta-blocker poisoning: a systematic review

Joe-Anthony Rotella^{a,b} , Shaun L. Greene^{a,c}, Zeff Koutsogiannis^{a,b}, Andis Gaudins^{a,d,e}, Yit Hung Leang^a, Kelvin Kuan^f, Helen Baxter^g , Elyssia Bourke^a and Anselm Wong^{a,e,h}

^aVictorian Poisons Information Centre, Austin Health, Victoria, Australia; ^bDepartment of Emergency Medicine, Northern Health, Epping, Victoria; ^cDepartment of Medicine, Faculty of Medicine, University of Melbourne, Victoria, Australia; ^dMonash Toxicology and Emergency Department, Monash Health, Victoria, Australia; ^eDepartment of Medicine, School of Clinical Sciences at Monash Health, Monash University, Victoria, Australia; ^fDepartment of Emergency Medicine, Changi General Hospital, Singapore, Singapore; ^gAustin Health Library, Austin Health, Victoria, Australia; ^hDepartment of Medicine and Radiology, Centre for Integrated Critical Care, Melbourne Medical School, University of Melbourne, Victoria, Australia

ABSTRACT

Introduction: Beta-adrenoreceptor antagonist (beta-blocker) poisoning is a common overdose which can lead to significant morbidity and mortality.

Objective: To evaluate the effects of treatments for beta-adrenoreceptor antagonist poisoning.

Methods: Searches were conducted across MEDLINE (1946–26 November 2019, Ovid); Embase (1974–26 November 2019, Ovid); and the Cochrane Central Register of Controlled Trials (CENTRAL, to 26 November 2019) utilising a combination of subject headings and free text. The search strategy identified 15,553 citations. Two reviewers screened titles and abstracts prior to selecting 141 articles (Kappa on articles included = 0.982, 95% CI 0.980–0.985). Primary outcomes included mortality and improvement in haemodynamic parameters (e.g., heart rate, blood pressure or a composite measure able to quantitate a haemodynamic response).

Results: The risk of bias was high for all interventions.

Gastric decontamination: Fifteen case reports described the administration of activated charcoal and five detailed the use of gastric lavage. As there was concurrent utilisation of multiple interventions, it was difficult to draw definitive conclusions regarding the relative contribution of these interventions to mortality or survival.

Catecholamines, inotropes and vasopressors: The use of catecholamines in treating beta-blocker toxicity was reported in 16 case reports, 3 case series and 2 animal studies. These agents most likely provided a survival benefit and improved haemodynamics.

Atropine: Multiple intravenous boluses of atropine were associated with improvement in heart rate and blood pressure in one case report.

Calcium: Intravenous calcium was associated with an improvement in haemodynamics in three out of six case reports but in association with multiple other therapies as well as in two animal studies.

High-dose insulin euglycaemic therapy: The use of this therapy was associated with mortality benefit in 10 case series. Two case reports showed clear haemodynamic improvement in a timeframe consistent with insulin administration (bolus then continuous infusion). Maintenance dosing ranged from 1 to 10 units/kg/h of insulin. However, it is unclear whether high-dose insulin euglycaemic therapy improved haemodynamic response above catecholamines and other inotropic agents in humans. Hypoglycaemia and hypokalemia were commonly observed adverse effects.

Glucagon: Glucagon was associated with minor improvements in haemodynamics through an increase in heart rate in two case series, nine case reports and five animal studies.

Methylthioninium chloride (methylene blue): Four case reports reported an association with improvement in haemodynamics following administration of methylene blue but in the setting of co-ingestion with amlodipine.

Intravenous lipid emulsion therapy: There was variable response to intravenous lipid emulsion therapy reported in 10 case series, 5 animal studies and 21 case reports.

Lignocaine: There were four case reports showing variable response to lignocaine in arrhythmias secondary to beta-blocker toxicity.

Other treatments: Fructose diphosphate, levosimendan and amrinone did not provide a mortality or significant haemodynamic benefit in three animal studies and nine case reports.

Veno-arterial extracorporeal membrane oxygenation: Veno-arterial extracorporeal membrane oxygenation was associated with improved survival in patients with severe cardiogenic shock or cardiac arrest in an observational study and four case series.

Dialysis: The evidence of four case reports suggest haemodialysis may assist in the management of massive overdose of specific water-soluble beta-blockers (e.g., atenolol) by improving elimination; however, a survival or haemodynamic benefit was not established.

Pacing: One case series and a single case report showed the utility of temporary overdrive cardiac pacing to prevent arrhythmias in sotalol toxicity.

ARTICLE HISTORY

Received 24 September 2019

Revised 24 February 2020

Accepted 27 March 2020

KEYWORDS

Antidotes; beta-blockers; beta-adrenoreceptor antagonists; cardiotoxicity; management; overdose; poisoning; toxicity; treatment

Conclusions: Catecholamines, vasopressors, high-dose insulin euglycaemic therapy and veno-arterial extracorporeal membrane oxygenation were associated with reduced mortality. However, it must be acknowledged that multiple treatments were often given simultaneously. Haemodynamic improvements in blood pressure and cardiac output were seen with the use of catecholamines, vasopressin and high-dose insulin euglycaemic therapy. Evidence for treatment recommendations is almost entirely drawn from very low- to low-quality studies and subject to bias. However, it is reasonable to have a graduated response to cardiovascular instability beginning with intravenous fluids, commencement of a single or a combination of catecholamine inotropes and vasopressors depending upon the type of haemodynamic compromise (bradycardia, left ventricular dysfunction, vasodilation). High-dose insulin euglycaemic therapy can be introduced as an adjunctive inotrope and lastly, more invasive methods such as veno-arterial extracorporeal membrane oxygenation should be considered in cases unresponsive to other therapies.

Introduction

Beta-adrenoreceptor antagonists, more commonly known as “beta-blockers,” are prescribed for a variety of medical conditions in both adult and paediatric populations worldwide. These include, but are not limited to hypertension, long-term prophylaxis for angina pectoris, suspected or definite myocardial infarction, stable cardiac failure, cardiac dysrhythmias, migraine and anxiety [1]. Given the wide range of indications for beta-blockers, as well as a relatively narrow therapeutic index in the case of sotalol for example, the potential for death after overdose (both intentional and accidental) is significant [2].

While significant morbidity and death would be anticipated as a consequence of cardiovascular shock secondary to beta-adrenoreceptor antagonist poisoning, the toxicity exerted by specific agents may be heterogeneous after overdose. Some beta-blockers have additional properties that make them more toxic after overdose. For example, seizures and sodium channel antagonism after propranolol overdose [3] or potassium channel antagonism resulting in QT-interval prolongation after sotalol overdose. As a class, beta-blockers were ranked seventh out of the top 25 substances associated with death reported to the American Poison Controls Centres in 2018 (122 cases, making up 3.9% of substances involved in fatalities) [4]. These data are more than likely an under-estimation of the true incidence of beta-blocker poisoning morbidity and mortality.

There are several therapies reported in the medical literature addressing the management of beta-blocker poisoning. The effect of these treatments on morbidity and mortality has not been well defined and utilisation of some treatments is controversial. In some instances, these treatments may even cause harm. As beta-blockers differ both with regards to receptor profile as well as pharmacokinetic characteristics (Table 1), there are implications as to the effectiveness (or ineffectiveness) a treatment modality may have in the treatment of specific beta-blocker toxicity (e.g., dialyzability). A critical review of the literature is needed to inform future evidence-based clinical practice guidelines for the management of beta-blocker poisoning and, thus, ensure patients receive safe, effective and consistent treatment.

Objective

The objective of this systematic review (registered in Prospero, review number CRD42019102788) is to evaluate the reported effects of treatments used for beta-blocker toxicity. This may inform the development of clinical guidelines

in accordance with the AGREE (Appraisal of Guideline Research and Evaluation) II statement [7] and GRADE (Grading of Recommendations Assessment, Development, and Evaluation) methodology [8].

Methods

The methodology adopted in this review was similar to that used previously to perform systematic reviews on treatments used in other forms of cardiovascular drug poisoning [9]. A search of the literature across several databases was undertaken. Studies were selected for inclusion in the review if they reported the effects of treatments for beta-blocker poisoning in either human or animal participants. Poisoning was defined as an “*exposure (...) causing or capable of causing toxicity, regardless of intent.*” [10]. Both isolated exposures and exposures in combination with other drugs were included if they met the remainder of the inclusion criteria.

Study types

Controlled trials, observational studies, case series (defined as two or more cases of beta-blocker toxicity reported together), animal studies, single case reports and abstracts from scientific meetings without date restriction, were included.

Interventions

Studies considered for inclusion pertained to intervention/s aimed at improving the primary and/or secondary outcomes. Interventions included pharmacological agents (such as inotropes and vasopressors) and mechanical therapies (such as dialysis) used in the management of beta-blocker toxicity.

Outcome measures

Primary outcomes include mortality (in hospital, or at the end of the experiment for animal studies) and improvement in haemodynamic parameters (e.g., heart rate, blood pressure or a composite measure able to quantitate a haemodynamic response).

Secondary outcomes include length of hospital stay, length of hospital stay in ICU and reported adverse effects of treatment. Included studies were required to address at least one of the primary or secondary outcomes.

Table 1. Pharmacokinetic properties of various beta-blockers [5,6].

Drug	Absorption (% of dose)	Bioavailability (% of dose)	Major 1st pass hepatic metabolism	Protein binding (%)	Lipid solubility (%)	Half-life (h)	Route of elimination	Active metabolites	Physiological properties
Acebutolol	>90	~40	Yes	25	Low	3–4	HM, biliary, RE	Yes	β1 blocker
Atenolol	~50	~40	No	<5	Low	6–9	RE	No	β1 blocker
Betaxolol	>90	~80	No	50	Low	15	HM	No	β1 blocker
Bisoprolol	=90	~88	No	~33	Low	10–12	RE (50% unchanged), HM	No	β1 blocker
Carvedilol	>90	~30	Yes	98	Moderate	6–10	HM, biliary	Yes	β1, β2, α1 blocker
Esmolol ^b	NA	NA	NA	55	Low	9 min	Erythrocytes	No	β1 blocker
Labetalol	>90	~33	Yes	~50	Moderate	5–8	HM	No	β1, β2, α1 blocker, ISA
Metoprolol	>90	~50	Yes	12	Moderate	3–4	HM	No	β1 blocker
Metoprolol LA	>90	65–70	Yes	12	Moderate	3–7	HM	No	β1 blocker
Nebivolol	>90	12–96	Yes	98	NA	8–27	HM	No	β1 blocker
Oxprenolol	~90	19–74	Yes	80	Moderate	2–3	HM	No	β1, β2 blocker, ISA
Propranolol	>90	30–70	Yes	93	High	3–4	HM	Yes	β1, β2 blocker, Sodium channel blockade
Sotalol	NA	~90	No	0	Low	9–10	RE	No	β1, β2 blocker, HERG potassium channel blockade
Timolol	>90	~75	Yes	~10	Low–moderate	4–5	RE (20% unchanged), HM	No	β1, β2

^aDetermined by the distribution ratio between octanol and water.

^bUltra-short-acting beta-blocker, only available in intravenous form.

LA: long-acting; NA: not applicable or no data; HM: hepatic metabolism; RE: renal excretion; ISA: intrinsic sympathomimetic activity.

Search strategy

The authors developed comprehensive search strategies to identify relevant studies pertaining to the management of beta-blocker poisoning. Searches were conducted, with the assistance of a librarian, across MEDLINE (1946–26 November 2019, Ovid); Embase (1974–26 November 2019, Ovid); and the Cochrane Central Register of Controlled Trials (CENTRAL) utilising a combination of subject headings and free-text including: adrenergic beta-antagonists, beta-blocker*, beta antagonist*, beta adrenergic blocker, beta adrenergic receptor block*, beta adrenoreceptor blocker*, or beta adrenoreceptor antagonist*; and poisoning, drug overdose, toxicity, intoxication, ingestion*, adverse effect*, or medication error*. Individual beta-blocker drug names were included in the search using wildcards to account for variations in spelling. See Appendix 1 for full search strategy.

There were no date or language restrictions applied. Removal of duplicate studies occurred prior to title and abstract screening. Subject matter experts were consulted for further relevant publications and authors scanned the reference lists of included studies and searched for on-going trials in the Australian New Zealand Clinical Trials Registry and ClinicalTrials.gov.

Two independent reviewers, blinded to authors and journal names, selected the studies based on the inclusion criteria. Disagreements were resolved by consensus following discussion or, when required, by a third reviewer.

Included studies were categorised by treatment modality and data extracted by investigators. For each included study, a reviewer independently extracted, study characteristics (year of publication, authors, and study design), subjects (number, inclusion/exclusion criteria, co-ingestions, type of animal studied where applicable, sample size calculation), treatment and control group characteristics, beta-blocker involved (type, dose, route, and form), treatment(s) provided, outcomes, and results. The data were then entered into a specific table according to the type of study.

Controlled trials, case series and animal studies are entered in Table 2. Case reports were entered in Appendix 2.

The search strategy identified 15,553 citations (Appendix 3). Two reviewers screened titles and abstracts prior to selecting 141 articles for extraction and analysis (Kappa on articles screening by two reviewers = 0.982, 95% CI 0.980–0.985).

Quality analysis was performed for all studies except case reports. The STROBE (STrengthening the Reporting of OBservational studies in Epidemiology) checklist [14] and the Thomas tool [15] were used for observational studies. The Institute of Health Economics tool for case series [16], and the ARRIVE (Animal Research: Reporting of *In Vivo* Experiments) guidelines [17] and a checklist adapted from the National Research Council of the National Academies (NRCNA) *Guidance for the Description of Animal Research in Scientific Publications* (Appendix 4) [18]. A meta-analysis was not performed due to heterogeneity of studies, interventions and outcomes.

Results

One hundred and forty-one articles were selected and extracted for analysis. There were no human randomised controlled trials. The articles consisted of two observational studies, 26 case series, 24 animal studies and 89 case reports. Of the animal studies analysed, 23 were *in vivo* animal studies (four porcine model, nine canine, five leporine, five murine (one mouse, four rat) and one was an *ex vivo* studies (one mouse). One study examined responses to the same intervention (IV propranolol and verapamil) versus control intervention in rabbit, mice and swine. Table 2 describes the results of interventions reported in all included studies [11–13,19–66] except case reports. Data extracted from case reports for each intervention are available online (Appendix 2). Each reviewer performed the quality assessment for articles allocated to their assigned methodology.

Table 2. Results of individual studies and risk of bias from interventions used for beta-blocker toxicity.

Source	Intervention	Participants	Outcomes	Methodological quality
Catecholamines, inotropes and vasopressors				
Case Series				
Cox et al. [11]	Vasopressors versus HIET	15 people with CCB/BB toxicity	Mortality: Benefit if norepinephrine drip or phenylephrine push-dose; worse if epinephrine push-dose; no difference with HIET	Institute of Health Economics tool for quality of case series and quality of reporting 8/20 (3 partial, 3 unclear)
Robinson et al. [12]	HIET ± vasopressors ± inotropes	9 people with BB only toxicity 3 people with CCB/BB toxicity	Worse outcome with HIET in BB only group	
Musselman et al. [13]	HIET versus Inotropy (glucagon/vasopressors)	5 people with BB only toxicity 7 people with CCB/BB toxicity	MAP: no difference LOS/ICU LOS/ventilator days: no difference	
Case reports				
- Survival reported in 14 cases [67,69,74,78,84,85,88–92,152–155] - Improvement in haemodynamics attributed to vasopressors reported in 10 cases [67,74,78,84,89–92,153–155] - No improvement following vasopressor administration reported in 5 cases [69,85,88,152,156] - Adverse effects: none reported				
Animal studies				
ARRIVE guidelines and Modified NRCNA				
Katzung et al. [57]	Group 1: High dose insulin (HDI) Group 2: High dose insulin + norepinephrine (HDI + NE) Group 3: Norepinephrine + epinephrine (NE + E). Glucose given to maintain 40–60 mg/dl. Potassium given to keep K > 2.8 mmol/l	15 pigs poisoned with IV propranolol 0.5 mg/kg bolus + 0.25 mg/kg/min infusion until HR × MAP 75% baseline, then infusion decreased to 0.125 mg/kg/min and 20 ml/kg saline bolus followed by 1.5 mL/kg/h Group 1 received HDI (10 U/kg/h), Group 2 received HDI (10 U/kg/h), Group 3 received NE (0.1 mcg/kg/min titrated up to max of 0.5 mcg/kg/min). Propranolol increased stepwise. When MAP < 55 mmHg, Group 1 received saline, Group 2 received NE (0.1 mcg/kg/min titrated up to max of 0.5 mcg/kg/min), Group 3 received E (0.1 mcg/kg/min titrated up to max of 0.5 mcg/kg/min) until death or 4 h Porcine model. Compared 2 previous studies: high-dose insulin (HDI) versus vasopressin/epinephrine (V/E) with HDI response study to compare V/E and placebo	Cerebral tissue oxygenation: Group 2 > Group 1. Group 3 had too rapid collapse to analyse. Mortality: Group 1 survived 1.9 h Group 2 survived 2.9 h. Group 3 survived 0.1 h	ARRIVE: 17/20 NRCNA: 6/16
Engelbrechtsen et al. [32]	Group 1: vasopressin/epinephrine Group 2: placebo Not head-to-head comparison but extrapolation from previous studies		MAP: V/E 75.5 mmHg, placebo 49 mmHg SVR: V/E 2046 dyne × s/cm ⁵ , placebo 1263 dyne × s/cm ⁵ Cardiac output: no difference Survival: 5/5 pigs died in V/E group within 90 min; 2/4 pigs died in placebo group with 360 min Mortality: all Group 2 survived 4 h. All Group 1 died within 90 min. CO in Group 2 increased throughout the 4 h but decreased in Group 1 until death	ARRIVE: 2/20 NRCNA: 4/16
Holger et al. [44]	Group 1: epinephrine 10 mcg/kg/min titrated up 10 mcg/kg/min every 10 min to 50 mcg/kg/min or baseline achieved + 0.0028 U/kg/min titrated every 10 min to 0.014 U/kg or baseline achieved.	10 pigs poisoned with IV propranolol 0.5 mg/kg bolus + 0.25 mg/kg/min infusion until HR × MAP 75% baseline, then infusion decreased to 0.125 mg/kg/min and 20 ml/kg saline bolus. Monitored for 4 h or death		ARRIVE: 16/20 NRCNA: 6/16

(continued)

Table 2. Continued.

Source	Intervention	Participants	Outcomes	Methodological quality
Holger et al. [32]	Group 2: insulin 2 U/kg/h increased by 2 U/kg/h every 10 min to 10 U/kg/h or baseline achieved. Glucose given if <60 mg/dl Group 1: vasopressin 0.0028–0.014 U/kg/min Group 2: glucagon 0.05 mg/kg bolus +0.15 mg/kg/h	16 pigs poisoned with IV propranolol 1 mg/kg bolus + 0.25 mg/kg/min infusion until 25% reduction in HR and MAP, then 20 ml/kg saline bolus. Monitored for 4 h or death	Mortality: 1/8 survived in Group 1; 0/8 survived in Group 2. No difference in MAP, SBP, HR, CO, pH, glucose. MAP and SBP higher in Group 1 in the first 1 h	ARRIVE: 13/20 NRCNA: 7/16
Atropine				ARRIVE guidelines and Modified NRCNA
Animal studies				
Rials and Tse [33]	Group 1: none Group 2: 1 mg/kg atropine 4 min after cardiac arrest	Open chest dogs anaesthetised with Nembutal treated with Digitoxin 1 mcg/kg/min till development of AV block junctional rhythm and wide R wave. Treated with 0.75 mg/kg propranolol 30 min after; cardiac arrest occurs 1 min after and ventricles paced every other minute at 60 beats/min	2 dogs in Group 1: no spontaneous beats in between pacing 3 dogs in Group 2: atropine initiated junctional beats capable of maintaining the dog	ARRIVE 11/20 NRCNA 4/16
Kandil et al. [22]	Group 1: IP propranolol 1 mg/100 g BW Group 2: IP atropine 500 mcg/100 g BW Group 3: propranolol then atropine 15 min after Group 4: propranolol and atropine simultaneously Group 5: atropine 15 min before propranolol Group 6: IP distilled water 0.2 ml × 2 as control	72 Albino mice anaesthetized with thiopental sodium; ECG tracing every 10 min for 90 min	Atropine is able to antagonise the effect of propranolol on the heart rate especially when administered simultaneously or before propranolol. The post propranolol injection of atropine had a weak correcting effect	ARRIVE 12/20 NRCNA 3/16
Case reports				
• Survival reported in 2 cases [85,92]				
• Improvement in haemodynamics (heart rate) following atropine reported in one case [69]				
Adverse effects: none reported				
Calcium				
Animal studies				ARRIVE guidelines and Modified NRCNA
Love et al. [37]	Anaesthetized mongrel dogs treated with IV 10% calcium chloride 0.7 mEq/kg versus IV normal saline solution	11 Dogs anaesthetized and given IV 10 mg/kg DL propranolol 1 dog died and 5 was assigned to each group	Calcium improved cardiac index, maximal ventricular pressure change over time and stroke volume	ARRIVE 17/20 NRCNA 8/16
Langemeijer et al. [26]	3 experiments: 1. Toxic dose–response relationship determination 2. Contractility assessment with ventricular pacing model 3. Infusion of calcium followed by beta-blocker administration After a 30 min stabilisation period, hearts were perfused with a high, normal or low calcium perfusion medium, followed by the same medium	Male rats anaesthetised and hearts prepared according to Langendorff procedure 3 groups for each drug [low ($n = 4$), normal ($n = 6$) and high ($n = 4$) calcium perfusate] comparing timolol, propranolol, sotalol	Low calcium perfusate with beta-blocker resulted in marked decline in myocardial contractility, whereas high calcium perfusate reduced the beta-blocker's contractility effect. There were no effects on heart rate and AV conduction	ARRIVE 17/20 NRCNA 9/16

(continued)

Table 2. Continued.

Source	Intervention	Participants	Outcomes	Methodological quality
Vick et al. [58]	with added beta-blocker (propranolol, sotalol or timolol), followed by a drug free calcium medium as before and finally with a normal calcium medium Rabbits: IV calcium chloride 300 mg/kg after verapamil infusion Miniature swine: 5 treated with IV calcium chloride 150 mg/kg after propranolol and verapamil injection Mice: pre-treatment with IV calcium chloride 600 mg/kg	30 rabbit controls: IV propranolol 1 mg/kg followed by IV infusion of verapamil 0.1 mg/kg/min for 15 min; 12 rabbits treated with intervention 8 swine controls under sodium pentobarbital anaesthesia: IV propranolol 0.5 mg/kg and verapamil 0.5 mg/kg; 5 swine treated with intervention 30 mice controls: intraperitoneal propranolol 13.6 mg/kg and verapamil 6.4 mg/kg	Calcium prevented death in all intervention animals. Intervention rabbits and swine had normalisation of blood pressure and cardiac conduction but persistent bradycardia	ARRIVE 17/20 NRCNA 8/16
Case reports				
- Survival reported in 6 cases [70,71,79,84,93,94] - Improvement in haemodynamics attributed to calcium reported in 3 cases [70,71,94] - Adverse effects: none reported				
High-dose insulin euglycaemic therapy (HIET)				
Case series				
Institute of Health Economics tool for quality of case series and quality of reporting				
Cole et al. [38]	HIET ≥ 1 unit/kg	88 patients with BB toxicity 45 patients with combined BB and CCB toxicity (66 patients with CCB toxicity) 22 patients 12 BB 6 CCB 3 BB + CCB 1 FWS bite	Mortality: 20/133 Cardiac arrest: 29/133 Hypoglycaemia: 46/133 Hypokalaemia: 33/133 Mortality: 3/22 Mild Hypoglycaemia (2.5–3.4 mmol/l or 45–62 mg/dl): 7/22 Severe Hypoglycaemia (<2.5 mmol/l or <45 mg/dl): 9/22 Neurological effects from hypoglycaemia: 0/22 Hypokalaemia: 18/22 No association between insulin dosing and severity of hypoglycaemia or hypokalaemia OR in replacement with glucose or potassium HIET group more likely to get multiple vasoactive therapies	14/20
Page et al. [61]	Patients receiving HIET ≥ 1 unit/kg for toxin-induced cardiac toxicity			17/20
Ryan et al. [19]	Review of clinical features and treatment of BB and CCB poisoning	27 patients 14 HIET (10 BB, 4 CCB) 13 catecholamines (9 BB, 4 CCB)		10/20 (3 unclear)
Robinson et al. [12]	Combination of HIET up to 10 units/kg prior to addition of vasopressors AND/OR inotropes	14 patients 9 BB 2 CCB 3 BB + CCB	Mortality: 3/14 Hypoglycaemia: 7/14 Hypokalaemia: 5/14 Pulmonary oedema: 3/14 Gangrenous digits: 1/14 Mortality: 6/8 R, 6/6 NR Response to HIET with BB: 62.5% (versus 16.7%) Response to HIET with BB + CCB: 25% (versus 33%) Hypoglycaemia: 9/14	12/20 (1 unclear)
Gerwin et al. [55]	HIET, glucagon and lipid emulsion in CCB and BB overdose	14 patients 8 Responders (R) 6 Non-responders (NR)		10/20 (1 unclear)

(continued)

Table 2. Continued.

Source	Intervention	Participants	Outcomes	Methodological quality
Riggan et al. [63]	Use of HIET ± IFE in CCB or BB toxicity	26 patients • 16 BB • 8 CCB 2 BB + CCB 215 exposures with CCB or BB toxicity	Mortality: 2/20 (both HIET + IFE)	5/20 (5 unclear)
Darracq et al. [64]	Assessment of how often HIET or IFE was recommended and how often recommendations were implemented Faxable sheet used as intervention to increase adherence to advice HIET ± IFE in CCB or BB toxicity	3 patients • 1 BB • 1 CCB • 1 CCB + TCA 12 patients • 5 BB • 2 CCB • 2 BB + CCB • 1 TCA 2 Poly-drug 20 patients (10 control, 10 HIET) • 8 CCB • 5 BB • 7 CCB + BB	HIET advice followed: 30/71 IFE advice followed: 10/30	9/20 (2 unclear)
Jacques et al. [27]			Improvement in haemodynamics ROSC 20 min after HIET + IFE given post cardiac arrest	3/20 (7 unclear)
Holger et al. [41]	HIET ≥ 1 unit/kg for toxin-induced cardiogenic shock		Mortality: 11/12 (polydrug) Hypoglycaemia: 6/12 Hypokalaemia: 8/12	8/20 (3 unclear)
Musselman et al. [13]	Patients receiving inotropic therapy AND HIET for BB or CCB toxicity versus those receiving inotropic therapy only		No significant difference in change of mean arterial blood pressure from baseline No significant difference in median hospital LOS, ICU LOS or ventilator days Median duration of inotropic therapy shorter in control than HIET group (7.8 versus 23 h)	9/20 (1 unclear)
Case reports				
• Survival reported in 17 cases [72–74,76,79,82,83,91,96,112,119,120,154,157–160] • Improvement in haemodynamics attributed to insulin reported in 11 cases [72–76,83,96,154,157,158,160] • No improvement reported following insulin administration in 3 cases [79,161] Adverse effects reported in 5 cases [76,82,96,157,159]				
Animal studies				
Katzung et al. [57]	Randomised control study comparing HDI to vasopressors or combination therapy on cerebral tissue oxygenation in refractory-propranolol-induced cardiogenic shock Group 1: HDI Group 2: HDI + Norepinephrine (NE) Group 3: Norepinephrine + Epinephrine (E) Treatment for propranolol-induced cardiogenic shock Group 1: Placebo Group 2: HIET 1 unit/kg Group 3: HIET 5 unit/kg Group 4: HIET 10 unit/kg	15 swine • 5 pigs in Group 1 • 5 pigs in Group 2 • 5 pigs in Group 3 16 pigs 4 pigs in each group	Mortality within 240 min (Group 1): 5/5 Mortality within 240 min (Group 2): 3/5 Mortality within 240 min (Group 3): 0/5 HDI + NE maintained cerebral tissue oxygenation more than HDI. Unable to assess NE + E due to rapid death	ARRIVE: 16/20 NRCNA: 6/16
Cole et al. [162]			Mortality (Group 1): 2/4 Mortality (Group 2): 1/4 Mortality (Group 3): 1/4 Mortality (Group 4): 0/4 CO increased corresponding with dose increase Rate pressure product (HR × MAP) increase greater with insulin/dextrose than ILE	ARRIVE: 19/20 NRCNA: 11/16
Harvey et al. [43]	Comparison of efficacy between insulin/dextrose and ILE in combined enteric/IV propranolol toxicity	10 adult New Zealand White Rabbits 5 rabbits in each group		ARRIVE: 16/20 NRCNA: 8/16

(continued)

Table 2. Continued.

Source	Intervention	Participants	Outcomes	Methodological quality
	Group 1: 3 units/kg insulin plus 0.5 g/kg glucose plus repeat dose if cardiovascular collapse Group 2: 10 ml/kg bolus ILE followed by further 3 ml/kg in event of cardiovascular collapse Crossover if no response to second dose Comparison of insulin versus glucagon and epinephrine in propranolol toxicity Group 1: control (no toxin, saline) Group 2: Sham treatment Group 3: insulin 4 units/min Group 4: glucagon 50 µg/kg bolus then 150 µg/kg/h Group 5: Epinephrine 1 µg/kg/min	27 mongrel dogs • 3 dogs in Group 1 • 6 dogs in Group 2 • 6 dogs in Group 3 • 6 dogs in Group 4 • 6 dogs in Group 5	Mortality (Group 1): 4/5 Mortality (Group 2): 2/5	
Kerns et al. [42]			Mortality (Group 1): 0/3 Mortality (Group 2): 6/6 Mortality (Group 3): 0/6 Mortality (Group 4): 4/6 Mortality (Group 5): 1/6	ARRIVE: 15/20 NRCNA: 11/16
Glucagon				
Case series				Institute Health of Economics tool for quality of case series and quality of reporting
Catlin et al., abstract [65]	Response to bolus of high-dose glucagon in suspected BB toxicity	Adult patients Participants dichotomized as good responders (10-min HR change greater than or equal to 10 bpm) and poor responders (10-min HR change less than 10 bpm)	Good response: 22/39 Poor response: 17/39 Median heart rate increased 12 bpm for 10 min post glucagon Median MAP increased 11 mmHg for 10 min post glucagon Good response was noted to correlate with lower actual body weight compared to poor response (77.8 versus 87.3 kg; $p = 0.044$, t test) Weight-based glucagon dosing was significantly higher for the good responders (63.9 versus 56.4 mcg/kg; $p = 0.036$, t test)	5/20
Gerwin et al. [55]	Patients admitted for presumed CCB and/or BB overdose were screened. Patients with a mean arterial pressure <65 mmHg that received HIET, glucagon, or lipid emulsion were included and grouped into responders (R) or non-responders (NR) based on hemodynamic improvement observed following therapy	14 patients • 8 Responders • 6 Non-responders	No differences in baseline demographics between groups (age 49 ± 21 versus 52 ± 15 year, $p = 0.82$; sequential organ failure assessment score 10 ± 4 versus 9 ± 5, $p = 0.63$; number of comorbidities 4 [1–5] versus 2 [2–3], $p = 0.57$). No differences in the agents overdosed on between R and NR were observed (CCB: 12.5% versus 50%, $p = 0.25$; BB: 62.5% versus 16.7%, $p = 0.14$; CCB & BB: 25% versus 33.3%, $p = 1.0$). Vasopressor use (50% versus 66.7%, $p = 0.63$) and survival (75% versus 100%, $p = 0.47$) were similar. Use of initial therapies were similar between groups (HIET: 25% versus 16.7%, $p = 1.0$; glucagon: 50% versus	4/20

(continued)

Table 2. Continued.

Source	Intervention	Participants	Outcomes	Methodological quality
Case reports				
- Survival reported in 11 cases [77–79,97–102,136,163] - Improvement in haemodynamics attributed to glucagon reported in 9 cases [77–79,97–102] - No improvement reported in 1 case [136] - Adverse effects reported in X cases [79,99,163]			33.3%, $p = 0.63$; lipids: 25% versus 50%, $p = 0.58$). Nine (66.7%) patients treated with HIET had a blood glucose <70 mg/dl	
Animal studies				
Love et al. [30]	Three groups Response following IV propranolol - Control saline - Glucagon 20 µg/kg bolus - Amrinone 4 mg/kg bolus Similar fluid loads	18 mongrel dogs Convenience sample 6 per group I&V sedated	Both high-dose amrinone and glucagon given as a single dose increased dP/dtmax, CO, SV to a similar degree compared to saline control but had no effect on MAP or SVR Glucagon increased HR but amrinone did not CO increased after milrinone and glucagon administration. Glucagon increased HR and caused tachycardia. Milrinone increased SV and CO and reduced CVP and PCWP. Suggests increased contractility. Glucagon mainly increased HR to increase CO without fall in CVP and PCWP	ARRIVE: 11/20 NRCNA: 6/16
Sato et al. [53]	Three groups Response following IV propranolol - Milrinone 300 µg/kg - Glucagon 20 µg/kg - Control saline group ANOVA analysis	13 mongrel dogs Unmatched numbers per group Convenience sample I&V sedated	Glucagon increased HR but amrinone did not CO increased after milrinone and glucagon administration. Glucagon increased HR and caused tachycardia. Milrinone increased SV and CO and reduced CVP and PCWP. Suggests increased contractility. Glucagon mainly increased HR to increase CO without fall in CVP and PCWP	ARRIVE: 10/20 NRCNA: 5/16
Sato et al. [29]	Four groups Response following IV propranolol - Saline - Glucagon 2.5 µg/kg - Milrinone 100 µg/kg - Milrinone + glucagon (same dose as above) ANOVA analysis	23 mongrel dogs 3 excluded as died before Rx Convenience sample 5 per group I&V sedated	Glucagon increased HR at this lower dose, but did not increase CO or SV to baseline values Milrinone increased CO and SV to baseline values but not HR compared to control M + G resulted in greatest increase in HR, even more than glucagon alone to tachycardic rates > 200 bpm compared to milrinone and control. M + G increased CO but not SV compared to control and M alone Outcome was survival at 4 h Glucagon: 4/4 died before 4 h Vasopressin: 3/4 vasopressin pig survived to 4 h Vasopressin significantly increased SBP and MAP in first hour. No diff in CO HR pH No control group to assess no treatment	ARRIVE: 10/20 NRCNA: 6/16
Holger et al. [32]	Two groups Propranolol 1 mg/kg then infusion 0.25 mg/kg/min – 4 hours max aiming for 25% drop in HR × MAP product then IV Nsaline 20 ml/kg prior to intervention - Vasopressin 0.0028–0.014 U/kg/min titrated up every 5 min - Glucagon 0.05 mg/kg then 0.15 mg/kg/h Aimed for BP 85% of baseline for vasopressin group	16 pigs 12-week old Tracheostomy, ventilated, inhalational anaesthetic 8 per group Randomised to treatment at peak toxicity No control group	Outcome was survival at 4 h Glucagon: 4/4 died before 4 h Vasopressin: 3/4 vasopressin pig survived to 4 h Vasopressin significantly increased SBP and MAP in first hour. No diff in CO HR pH No control group to assess no treatment	ARRIVE 15/20 NRCNA 7/16

(continued)

Table 2. Continued.

Source	Intervention	Participants	Outcomes	Methodological quality
Kosinski et al. [36]	Glucagon 1 mg bolus v Isoprenaline 20 mg bolus Pre- and post-administration of 2 mg propranolol IV	Mongrel canines $\times$ 8 Tracheostomy, ventilated, pentobarb anaesthesia. Invasive aortic BP and coronary flow measure Invasive LV pressure measure	Before propranolol: Isoprenaline: increased heart rate, myocardial contractility and decreased MAP and coronary resistance. Glucagon: increased HR, MC contractility and MAP but did not change coronary resistance After propranolol: Isoprenaline: no effect on the above parameters Glucagon: persistence of all effects noted above	ARRIVE 9/20 NRCNA 1/16
Intravenous lipid emulsion (ILE)				
<i>Observational studies</i>				
Smolinske et al. [66]	ILE in addition to Conventional therapy	102/459(22%)	5(4.9%) with ROSC	<i>Strobe and Thomas Tool</i> STROBE: 19/22 Thomas tool: • Moderate: selection bias, study design, cofounders, Weak: intervention integrity, data collection methods, blinding, withdrawals <i>Institute of Health Economics tool for quality of case series and quality of reporting</i> 3/20
<i>Case series</i>				
Gunaydin et al. [49]	AC, fluids, catecholamines, glucagon, 1.5 ml/kg 20% ILE followed by 0.25 ml/kg/min	5 patients • 2 with BB poisoning	Mortality: 2/5 (both BB cases survive) Pleural effusion, pancreatitis: 1/5	
Jovic-Stosic et al., abstract [20]	Patients with poisoning receiving ILE Limited information in abstract (not subsequently published)	31 patients with poisoning receiving ILE • 7 with BB poisoning	Improved observed in most patients after ILE administration "increased BP and reversal of wide complex tachycardia are the most impressive effect" Mortality: 11/36 Small increase in MAP but not statistically significant. Interference with lab analyses: 6/36 No systemic data collection of adverse effect Mortality: 1/6 (BB)	Insufficient information to calculate
Mithani et al. [28]	Variable doses. From 1–2 ml/kg to boluses of 50–400 ml	36 patients • 10 with BB poisoning		8/20
Sebe et al. [60]	ILE 20% 1.5 ml/kg in 5 min, repeated if remains haemodynamically unstable/ poor perfusion after 15 min. Maintenance 0.25 ml/kg/h Patient 1: calcium, glucagon, HDI, 100 ml 20% ILE, catecholamines (pacing, IABP) Patient 2: calcium, HDI, catecholamines, 150 ml 20% ILE (pacing, IABP)	15 patients • 6 with BB toxicity Two patients Patient 1 • 50 mg Metoprolol, 150 mg bupropion XL (total 80 tabs) Patient 2 • 3.6 g diltiazem SR, 1.2 g propranolol		11/20
Cole et al. [21]			Mortality: 2/2 Both died shortly after ILE	4/20 (1 unclear)
Doepker et al. [40]	1.5 ml/kg followed by 0.25 ml/kg/min for 60 min Patient 1: atropine, glucagon, calcium, catecholamines, HDI, ILE	Two patients Patient 1 • 150 mg amlodipine, 10.26 g verapamil, 1.18 g metoprolol	Mortality: 0/2 Non-fatal hypoglycaemia in both cases	3/20

(continued)

Table 2. Continued.

Source	Intervention	Participants	Outcomes	Methodological quality
Cevik et al. [45]	Patient 2: AC, calcium, dextrose, glucagon, NaHCO ₃ , catecholamines, HDI, ILE 100 ml bolus then 0.5 ml/kg/min for 120 min	Patient 2 • 300 mg amlodipine, simvastatin, 300 mg lisinopril, 30 g metformin 10 patients • 1 with BB toxicity	Mortality: 1/10 (BB patient survived) BB patient survived, others unknown	6/20 (1 unclear)
Jacques et al. [27]	Glucagon, calcium, catecholamines, HIET, ILE, pacing	Three patients 1 with BB toxicity		Insufficient information to calculate
Tabone et al. [23]	Bolus 100–150 ml ILE No clear dose by weight Some received infusion and some did not with variation in duration of infusion	Combination of eight case reports and one case series of two patients	Mortality: 1/10 (2 unclear outcome)	Insufficient information to calculate
Meehan et al. [24]	Glucagon, HIET, catecholamines, ILE (dose not reported)	Two patients with BB toxicity	Mortality: 0/2	Insufficient information to calculate
Case reports				
• Survival reported in 16 cases [72–74,91,108–112,114–116,122]				
• Improvement in haemodynamics attributed to ILE implied in the above cases [72–74,91,108–112,114–116,122]				
• Adverse effects reported in 2 cases [113,118]				
Animal studies				ARRIVE guidelines and Modified NRCNA
Harvey et al. [43]	Group 1: 10 ml/kg 20% ILE over 5 min Group 2: 3 unit/kg insulin, 0.5 g/kg glucose and 7 ml/kg 0.9% saline solution over 5 min	10 NZ white rabbit underwent mini-laparotomy and enterostomy with 40 mg/kg propranolol instilled into the proximal small bowel. At 30 min, propranolol 0.3 mg/kg was injected intravenously and thereafter continued at 4 mg/kg/h until MAP 50% of baseline MAP	High-dose insulin resulted in greater haemodynamic recovery. At termination of study protocol, 1/5 from insulin-treated group and 3/5 from the ILE group were undergoing resuscitation treatment	ARRIVE: 18/20 NRCNA: 5/16
Browne et al. [48]	Group 1: 6 ml/kg 20% ILE Group 2: 6 ml/kg saline solution	20 adult NZ white rabbits underwent metoprolol infusion 20 mg/ml at 20 ml/h to MAP 60% of baseline MAP	No statistically significant difference in MAP or HR in both arms	ARRIVE: 14/20 NRCNA: 5/16
Cave et al. [52]	Group 1: 6 ml/kg 20% ILE Group 2: 6 ml/kg 0.9% saline solution	20 rabbits underwent atenolol infusion 25 mg/ml at 1 ml/min	Prior to intervention: Group 1 mortality: 4/10 Group 2 mortality: 2/10 After intervention: Group 1 mortality: 0/6 Group 2 mortality: 2/8 Greater increment in MAP observed in Group 1 but not statistically significant	ARRIVE: 7/20 NRCNA: 4/16
Harvey et al. [46]	Group 1: 6 ml/kg 20% ILE Group 2: 6 ml/kg 0.9% saline	20 NZ white rabbits underwent propranolol infusion 4.2 mg/min to MAP 60% of baseline MAP	MAP greater in ILE-treated group at 15 min. A trend toward increased MAP in ILE group observed at 12.5 min. No statistically significant in first derivative of MAP between groups	ARRIVE: 14/20 NRCNA: 5/16
Cave et al. [59]	Group 1: pre-treated with 16 ml/kg 20% ILE Group 2: pre-treated with 16 ml/kg 0.9% NaCl	10 <i>Rattus norvegicus</i> rats pre-treated with ILE or NaCl had constant infusion of propranolol until death	Lethal dose is higher in Group 1 but is not statistically significant. % of QRS change was reduced in Group 1	ARRIVE: 6/20 NRCNA: 1/16
Other treatments				

(continued)

(continued)

Table 2. Continued.

Source	Intervention	Participants	Outcomes	Methodological quality
<i>Animal studies</i>				
Kalam et al. [47]	Group 1 – IV propranolol + IV glucose Group 2 – IV propranolol + 10% 1,6 fructose-diphosphate 125 mg/kg loading dose and infusion + IV glucose Group 3 – IV propranolol + 10% 1,6 fructose-diphosphate 250 mg/kg loading dose and infusion + IV glucose Group 1: 0.9% saline Group 2: levosimendan 36 µg/kg loading + 0.6 µg/kg/min Group 3: levosimendan 0.6 µg/kg/min Group 4: epinephrine 0.5 µg/kg/min Group 5: levosimendan 70 µg/kg loading + 0.12 µg/kg/min Group 6: levosimendan 1.2 µg/kg/min Group 7: levosimendan 70 µg/kg loading	20 Wistar rats IV propranolol infusion, interventions administered once BP dropped to 50% baseline	Time to death primary outcome Both doses of FDP lead to significantly longer survival compared to controls 250 mg/kg FDP statistically higher mean and systolic BP 125 mg/kg trended higher, but not statistically significant	ARRIVE guidelines and Modified NRCNA ARRIVE:9/20 NRCNA: 5/16
Kalam et al. [50]	Group 1: 0.9% saline Group 2: levosimendan 36 µg/kg loading + 0.6 µg/kg/min Group 3: levosimendan 0.6 µg/kg/min Group 4: epinephrine 0.5 µg/kg/min Group 5: levosimendan 70 µg/kg loading + 0.12 µg/kg/min Group 6: levosimendan 1.2 µg/kg/min Group 7: levosimendan 70 µg/kg loading	10 male Wistar rats per group Propranolol infused 24 mg/kg/h until MAP fell to 50%, then study interventions commenced. All animals received same amount of fluid	No difference baseline variables No difference in survival in any group compared to controls statistically significant No difference in CO – compared to control Levosimendan worsened BP in all groups compared to control. Epinephrine statistically significant increase in BP No differences in HR or QRS duration Survival: Levosimendan statistically significant versus control, no difference within groups CO: Statistically significant increase levosimendan over control and epinephrine MAP: no increase with levosimendan. Epinephrine statistically significant increase	ARRIVE:17/20 NRCNA: 5/16
Kalam et al. [51]	Group 1: 0.9% saline Group 2: levosimendan 36 µg/kg loading + 0.6 µg/kg infusion Group 3: levosimendan 0.6 µg/kg infusion Group 4: epinephrine 0.5 µg/kg/min	10 male Wistar rats per group Metoprolol infused 24 mg/kg/h until MAP fell to 50%, then study interventions commenced. All animals received same amount of fluid	No differences in HR or QRS duration Survival: Levosimendan statistically significant versus control, no difference within groups CO: Statistically significant increase levosimendan over control and epinephrine MAP: no increase with levosimendan. Epinephrine statistically significant increase	ARRIVE:17/20 NRCNA: 6/16
Sato et al. [29]	Group 1 n = 5 – IV propranolol + N/saline Group 2 n = 5 – IV propranolol + N/saline + milrinone Group 3 n = 5 – IV propranolol + N/saline + glucagon Group 4 n = 5 – IV propranolol + N/saline + glucagon + milrinone Group 1 n = 3 – IV propranolol + N/saline Group 2 n = 5 – IV propranolol + N/saline + glucagon Group 3 n = 5 – IV propranolol + N/saline + milrinone Group 1: Control group – IV propranolol + N/saline Group 2: Glucagon – IV propranolol + N/saline + glucagon Group 3: IV propranolol + N/saline + amrinone	20 mongrel dogs Administered IV propranolol 10 min in all study groups, then bolus saline, glucagon, milrinone or milrinone + glucagon – CVS status studied over following 30 min	HR – no difference Milrinone restored CO Milrinone + glucagon restored CO, but produced tachycardia Glucagon alone only produced tachycardia, no increase in CO	ARRIVE:13/20 NRCNA: 6/16
Sato et al. [53]	Group 1 n = 3 – IV propranolol + N/saline Group 2 n = 5 – IV propranolol + N/saline + glucagon Group 3 n = 5 – IV propranolol + N/saline + milrinone Group 1: Control group – IV propranolol + N/saline Group 2: Glucagon – IV propranolol + N/saline + glucagon Group 3: IV propranolol + N/saline + amrinone	13 mongrel dogs Administered IV propranolol 10 min in all study groups, then bolus saline, glucagon or milrinone – CVS status studied over following 30 min	Milrinone lead to significant increase in CO c.f. controls Glucagon: increase in HR, but less increase in CO (remained below baseline). Milrinone: no effect on HR Compared to controls, IV glucagon increased CO at 1, 6, and 11 min post-administration + IV amrinone increased CO at 1-min post-administration. Glucagon had positive chronotropic effect. Amrinone had no positive chronotropic effect	ARRIVE:13/20 NRCNA: 5/16
Love et al. [30]	Group 1: Control group – IV propranolol + N/saline Group 2: Glucagon – IV propranolol + N/saline + glucagon Group 3: IV propranolol + N/saline + amrinone	18 mongrel dogs Three groups of 6 Administered IV propranolol + N/saline to all. Study groups given IV bolus of glucagon, or IV bolus amrinone	Milrinone lead to significant increase in CO c.f. controls Glucagon: increase in HR, but less increase in CO (remained below baseline). Milrinone: no effect on HR Compared to controls, IV glucagon increased CO at 1, 6, and 11 min post-administration + IV amrinone increased CO at 1-min post-administration. Glucagon had positive chronotropic effect. Amrinone had no positive chronotropic effect	ARRIVE:16/20 NRCNA: 5/16

(continued)

Table 2. Continued.

Source	Intervention	Participants	Outcomes	Methodological quality
Andersson et al. [56]	Group 1: Control group $n = 3$ metoprolol IV boluses + infusion Group 2: Test group $n = 4$ metoprolol IV boluses + infusion, then prenalterol IV boluses	Seven dogs • Given metoprolol boluses + infusion • After 3 h, four dogs given IV prenalterol boluses	Group 1 mortality: 3/3 Group 2 mortality: 0/4	ARRIVE: 9/20 NRCNA: 4/16
Kosinski et al. [36]	Groups not clearly described. Doses of interventions not clearly described. Glucagon 1 mg, then propranolol 20 mg/kg in some animals. Isoproterenol and then propranolol 20 mg/kg other animals	8 mongrel dogs. CVS response to glucagon or isoproterenol measured, then propranolol given and glucagon and isoproterenol given again	After propranolol, isoproterenol had no effect on HR, LV contractility, coronary artery flow or resistance. Glucagon did however positively affect these. However, no statistics done to make a comparison	ARRIVE: 5/20 NRCNA: 3/16
Extracorporeal membrane oxygenation (ECMO) Observational studies Masson et al. [31]	ECMO	62 poisoning or cardiac arrests (6 beta-blockers)	Mortality: In cardiac arrest 3/3 survival with ECMO versus 0/7 without ECMO. In severe shock 9/11 survival with ECMO versus 23/41 without ECMO. Adverse effects: 2 cases bleeding, 1 inferior vena cava thrombus, 4 limb ischaemia	STROKE and Thomas Tool STROBE: 17/22 Thomas tool: • Strong: Selection bias • Moderate: cohort methods, blinding, withdrawal, confounding Weak: intervention integrity
Case series				Institute of Health Economics tool for quality of case series and quality of reporting 12/20 (2 unclear)
Pozzi et al. [39]	ECMO	12 patients with either cardiac arrest or in shock (4 beta-blockers)	Mortality: 3/12 deaths Function: All 9 survivors had no neurological sequel. 75% survival at 1 year follow up Adverse effects: 1 ischaemic stroke, 2 iliofemoral thrombectomy, 3 compartment syndrome, 1 amputation	13/20 (2 unclear)
Daubin et al. [35]	ECMO	17 patients with severe cardiac toxicity or arrest (9 beta-blockers)	Mortality: 4/17 deaths Function: 13/17 survival with no cardiovascular or neurological sequel Adverse effects: 6 limb ischaemia, 1 IVC thrombus, 1 femoral thrombus, 2 bleeding from cannula site	15/20 (3 unclear)
Megarbane et al. [34]	ECMO	12 out of hospital cardiac arrest from poisoning (4 beta-blockers)	Mortality: 4/12 survived to ICU discharge (3 of these took beta-blockers) Function: 3 without complications at 1 year. Adverse: Nil	9/20 (9 unclear)
Babatasi et al. [62]	ECMO	6 cardiotoxic arrests	Mortality: 2/6 deaths Adverse effects: Limb ischaemia 3, retroperitoneal haematoma 1	
Case reports				
• Survival reported in 10 cases [81,112,143–147,164–166] • Improvement in haemodynamics attributed to ECMO in all 11 cases [80,81,112,143–147,164–166] • Adverse effects reported in 4 cases [80,143,164,165]				
Pacing Case series				
Totterman et al. [54]	Overdrive transcutaneous cardiac pacing [104]	4 patients with sotalol toxicity	Mortality: 0/4	Institute of Health Economics tool for quality of case series and quality of reporting 11/20

BB: beta-blocker; CCB: calcium channel blocker; ECMO: extracorporeal membranous oxygenation; CVVHDF: continuous venovenous haemodiafiltration; SDAC: single-dose activated charcoal; CPR: cardiopulmonary resuscitation; ALS: advanced life support; ILE: intravenous lipid emulsion; MAP: mean arterial blood pressure; U: units; LOS: length of stay; HIET: high-dose insulin euglycaemic therapy (1–10 units/kg bolus ± infusion) or HDI: high-dose insulin (1–10 units/kg bolus ± infusion); CO: cardiac output; HR: heart; CVP: central venous pressure; PCWP: pulmonary capillary wedge pressure; NE: norepinephrine or noradrenaline; E: epinephrine or adrenaline.

The following paragraphs summarise the main findings from the review.

Results: medical interventions

Gastric decontamination

Fifteen case reports [67–81] described the administration of activated charcoal. Seven studies reported the dose of activated charcoal (median dose 50 g, range 50–150 g) [68,69,72,74,79,82,83]. Of these cases, six involved atenolol, three involved bisoprolol, three involved propranolol, two involved metoprolol and two involved carvedilol. Death occurred in two cases as a result of progression of beta-blocker toxicity [75,80]. Five case reports [68–71,84] reported survival without complication following use of gastric lavage. One case report [85] described survival following copious vomiting of pill fragments induced by syrup of ipecac administration. Heise et al. [86] reported patient survival after removal of pill fragments by oesophageal gastroduodenoscopy 16 h post-ingestion of atenolol 4500 mg, coingested with chlorthalidone and lisinopril. Death resulted in one case following gastric lavage and administration of activated charcoal [69]. This was most likely the result of progression of beta-blocker toxicity following development of cardiogenic shock and pulmonary oedema.

Catecholamines, inotropes and vasopressors

The use of catecholamines in treating beta-blocker toxicity is reported in 16 case reports, 3 case series and 2 animal studies. In the three case series reporting use of catecholamines [11–13], there were 39 patients in total, of whom 14 ingested beta-blocker alone (36%). Musselman et al. [13] reported no difference between high-dose insulin euglycaemic therapy and glucagon/vasopressors (catecholamine or non-catecholamine) combination in five patients with regards to mean arterial pressure (56 ± 18 mmHg control versus 47 ± 18 high-dose insulin euglycaemic therapy mmHg), length of hospital stay, length of hospital stay or days requiring ventilation. Specific agents were not mentioned.

Robinson et al. [32] reported a series of 14 patients, of whom 9 had ingested a beta-blocker alone. Administration of high-dose insulin euglycaemic therapy alone as a rescue therapy resulted in less favourable outcomes excluding mortality but the details of which parameters were affected could not be elucidated from the available literature. The most common catecholamine used in this series of fourteen patients was norepinephrine ($n = 9$). Other vasopressors/inotropes used were phenylephrine ($n = 2$), vasopressin ($n = 5$), epinephrine ($n = 2$), and methylthioninium chloride ($n = 1$). There was no mention which inotropes/vasopressors were used for the beta-blocker only poisonings.

Of the 16 case reports, 7 described severe propranolol toxicity that was treated with multiple catecholamines and multiple other interventions (including intubation, non-catecholamine inotropes and vasopressors, cardiopulmonary resuscitation, extracorporeal life support, dialysis) were used

in severely unwell patients [69,85,87–91]. It is difficult to assess the individual effect of any specific catecholamine in these case reports.

Holger et al. [44] studied 10 pigs poisoned with IV propranolol (0.5 mg/kg bolus and 0.125 mg/kg/min infusion) subsequently randomised to either high-dose insulin euglycaemic therapy (insulin 2 units/kg/h increased by 2 units/kg/h every 10 min to 10 units/kg/h) or a vasopressin/epinephrine combination (epinephrine 10 mcg/kg/min titrated up 10 µg/kg/min every 10 min to 50 µg/kg/min or baseline achieved + 0.0028 units/kg/min titrated every 10 min to 0.014 units/kg or baseline achieved). The high-dose insulin euglycaemic therapy group was observed to have a significantly greater increase in cardiac output and a mortality benefit compared to the vasopressin/epinephrine combination. There was no placebo group and the study was not blinded.

Katzung et al. [57] reported a study of 15 pigs poisoned with IV propranolol and treated with either high-dose insulin euglycaemic therapy in combination with norepinephrine, high-dose insulin euglycaemic therapy alone or norepinephrine/epinephrine combination. The combination of high-dose insulin euglycaemic therapy with norepinephrine (mean survival time 2.9 h) was superior to high-dose insulin alone (mean survival time 1.9 h). In addition, both high-dose insulin groups were significantly superior to norepinephrine/epinephrine combination in terms of cerebral tissue oxygenation.

With regard to vasopressin use in beta-blocker toxicity, the evidence is limited to animal studies. In a porcine model, 16 pigs were poisoned with IV propranolol and randomised to either receive glucagon or vasopressin intravenously [32]. There was no difference in survival between groups (4-h Kaplan–Meier survival curves log-rank test 0.059, $p = 0.81$), but vasopressin infusion resulted in a higher mean arterial pressure and systolic blood pressure in the first hour of resuscitation. No control group was included in the study.

Atropine

Only animal studies and case reports were identified regarding the use of atropine in beta-blocker poisoning. Three case reports reported use of atropine [68,85,92] but only one reported haemodynamic improvement attributed to atropine use in the case of a 15-year-old female who took atenolol 500 mg [68]. Her heart rate improved from 44 to 60 post atropine 0.5 mg with a corresponding rise in blood pressure from 90/60 mmHg to 132/65 mmHg. She subsequently required eleven more doses during her admission to maintain her heart rate above 50 and her blood pressure above 90/60 mmHg. Similarly, two animal studies [22,33] suggested that atropine antagonises the bradyarrhythmic effects of propranolol.

Calcium

Only case reports and animal studies were identified reporting calcium use in beta-blocker toxicity. Of six case reports

[70,71,79,84,93,94], three reported haemodynamic improvement after calcium administration in three out of six patients [70,71,94]. Two cases reported improvement in blood pressure following 1 g and 2 g of intravenous calcium chloride, respectively, in patients who had taken an unknown amount of atenolol and presented with cardiogenic shock [70]. However, these cases also had co-ingestion of verapamil. A third case reported improvement in blood pressure following three doses of calcium chloride (1 g) with subsequent infusion in a patient presenting following an overdose of 1.8–2.5 g of atenolol [71]. There were no adverse effects reported that could be linked to calcium administration.

Two animal studies reported an improvement in cardiac index [37] and normalisation of blood pressure and cardiac conduction [58] following treatment with calcium chloride after administration of IV propranolol. One study [58] observed a cross-species mortality benefit from calcium chloride treatment in mice, swine and rabbits (all given propranolol and verapamil) compared to no treatment.

High-dose insulin euglycaemic therapy

There were 10 case series, 4 animal studies and 10 case reports identified reporting the use of insulin in beta-blocker poisoning; however, no controlled human trials or observational studies were identified. Each case series included isolated beta-blocker poisoning, isolated calcium channel blocker poisoning or combined beta-blocker and calcium channel blocker poisoning. It was not possible to isolate the outcomes of patients presenting with lone beta-blocker toxicity. Data were commonly pooled or incomplete across drug classes.

There was no significant difference observed in mortality between patients with beta-blocker poisoning treated with high-dose insulin euglycaemic therapy when compared to patients with either calcium channel blocker alone or combined calcium channel blocker and beta-blocker poisoning [38,55,95]. Two case series were derived from retrospective chart reviews of eligible patients at a single centre, with relatively small numbers in each (14 and 88, respectively) [38,55]. No significant difference in mean arterial pressure (from baseline) was observed in patients receiving high-dose insulin euglycaemic therapy in addition to vasopressors compared to vasopressors alone in a case series of 20 patients and the median duration of inotropic therapy was shorter in the control group [13]. The same study found no significant difference in median hospital length of stay, ICU length of stay or ventilator days [13].

Four case series specifically reported treatment with high-dose insulin therapy delivered as bolus dose of insulin followed by continuous infusion [27,38,41,61]. High-dose insulin therapy was variable in dose, ranging from 1 unit/kg/h up to 10 units/kg/h administered. There was limited information contained in the reports with respect to haemodynamic improvement.

While most case reports described an association between the administration of high-dose insulin euglycaemic therapy and improvement in haemodynamics (and in some cases,

survival), the responses were confounded by the use of multiple interventions with unclear timelines with regards to other inotrope or pressor administration and response. Two case reports showed a clear haemodynamic improvement in a timeframe consistent with insulin administration, in both instances a bolus was given followed by a continuous infusion— one case following an ingestion of 5 g of metoprolol and the other following an ingestion of an unknown amount of carvedilol [75,96].

Notably, adverse effects such as hypoglycaemia and hypokalaemia were commonly observed in five case series (42.6%, 195 cases) [12,41,55,61,95]. One case series also reported that the dose of insulin had no association with the severity of hypoglycaemia or hypokalaemia nor did it correlate with the amount of glucose or potassium supplementation required to correct said sequel [61].

Katzung et al. [57] reported improvement in porcine cerebral tissue perfusion following treatment with high-dose insulin euglycaemic therapy but no difference in eventual mortality. One study of 16 pigs appeared to support the use of high dose insulin with regards to survival following induction of propranolol-induced cardiogenic shock [25]. All pigs ($n = 4$) receiving 10 units/kg/h insulin survived in this study.

Glucagon

There were two case series, five animal studies and nine case reports pertaining to glucagon use. One case series examining the use of high-dose glucagon found that patients receiving a higher dose (10 mg) were more likely to have a haemodynamic response (22/39 patients). Median heart rate and mean arterial blood pressure increased by 12 beats per minute and 11 mmHg, respectively, post-dose [65]. However, when compared to high-dose insulin euglycaemic therapy and intravenous lipid emulsion for the treatment of beta-blocker toxicity, there was no difference in survival [55].

Nine case reports [77–79,97–102] described improvement in blood pressure but were highly variable with regards to timing and duration of effect in addition to being confounded by multiple treatments being utilised concurrently. Timing of administration in relation to effect was often poorly documented. One case report documented an improvement in haemodynamics (40–60 mmHg mean arterial pressure) 2 h after the commencement of a glucagon infusion in addition to dopamine and isoprenaline following an ingestion of 8 g of oxprenolol, but no response to a total of 30 mg IV glucagon boluses [100]. Another case reported improvement in haemodynamics (systolic BP 70–110 mmHg and heart rate 72–96/min) following two intravenous boluses of glucagon 10 mg after an overdose of propranolol 3 g in a 28-year-old male, who was also on an isoprenaline infusion [103]. Adverse effects reported included nausea and vomiting in two instances [79,98].

There was no mortality benefit in one animal study comparing glucagon (given at 0.05 mg/kg followed by an infusion of 0.15 mg/kg/h) against vasopressin in sixteen 12-week-old pigs after being administered propranolol 1 mg/kg followed by an infusion of 0.25 mg/kg/min [32]. All four pigs

given glucagon died within 4 h (the outcome being survival at 4 h). Animal studies all demonstrated that glucagon increased heart rate as a single agent but this did not cause significant changes in mean arterial blood pressure or cardiac output [29,30,32,36,53]. Two studies using mongrel dogs reported that 20 µg/kg glucagon given intravenously increased heart rate but did not affect mean arterial pressure (MAP) or systemic vascular resistance (SVR) when compared to amrinone (or milrinone) or saline control [30,53]. However when a lower dose of glucagon was used in a later study using 23 mongrel dogs, glucagon only increased heart rate but did not affect cardiac output or SVR [29].

Methylthioninium chloride

There were four case reports describing the use of methylthioninium chloride (methylene blue) in treatment of beta-blocker toxicity [104–107]. All four were cases reporting overdoses of atenolol in combination with amlodipine. In each case, methylthioninium chloride was used as a rescue therapy for vasodilatory shock caused primarily by amlodipine refractory to vasopressors with an improvement in haemodynamics.

Intravenous lipid emulsion therapy

To date, there are no controlled clinical studies that assess the effects of intravenous lipid emulsion in beta-blocker poisoning. There was 1 observational study, 10 case series, 5 animal studies and 21 case reports. In an observational study by Smolinske et al. [66], out of 102 overdoses involving beta-blocker toxicity and lipid therapy, only 5 (4.9%) had return of spontaneous circulation post arrest. One case series of 36 patients receiving intravenous lipid emulsion (10/36 with beta-blocker poisoning) did not show a clinical improvement in mean arterial blood pressure post lipid emulsion therapy [28]. Eleven patients died (three deaths were from a cardio-toxin – although, the authors did not define by drug class).

In the 21 published case reports, 13 describe some positive effect from intravenous lipid emulsion therapy (e.g., increased BP or heart rate) [72–74,91,108–124]. Thirteen of the reported cases were polydrug overdoses [73,74,91,110,113–121,123]. These all had multiple treatment interventions prior to intravenous lipid emulsion therapy and the actual response to intravenous lipid emulsion cannot be assessed.

Five animal studies report a trend to benefit with intravenous lipid emulsion therapy compared to placebo [43,46,48,52,59]. However, none were statistically significant. In one study, high-dose insulin therapy (3 U/kg with 0.5 g/kg glucose) was superior to intravenous lipid emulsion (20% Intralipid 10 mL/kg) in improving haemodynamics (heart rate and blood pressure) in ten adult New Zealand white rabbits after administration of IV propranolol 4 mg/kg/h [43].

Complications from intravenous lipid emulsion included interference with extracorporeal circuits (ECMO, dialysis) and laboratory investigations [113,117,118,125].

Lignocaine

Four case reports described the use of lignocaine to treat ventricular arrhythmias associated with sotalol toxicity [88,126–128]. In one case of sotalol poisoning (3 grams) in a 70-year-old male, torsades des pointes resolved after administration of a single dose of 100 mg of lignocaine. A further episode of torsades des pointes resolved after a further intravenous dose of lignocaine 50 mg. A lignocaine infusion of 3 mg/min was commenced. No further ventricular arrhythmias were observed over the following 36 h. The duration of the lignocaine infusion was not reported. [126]. In another case, ventricular tachycardia was terminated after administration of lignocaine (dose and route of administration were not reported) in 57-year-old female following the ingestion of sotalol 2.5 g [127]. Conversely, lignocaine failed to reverse ventricular tachycardia in a 25-year-old female following a sotalol 15 g ingestion [128].

Other treatments

There were nine case reports describing other interventions (enoximone, sodium bicarbonate, amrinone, prenalterol) for beta-blocker toxicity, however all provided minimal to no evidence of benefit [129–137].

Three animal studies have examined the effect of glucagon and a phosphodiesterase inhibitor (milrinone or amrinone) in treating propranolol toxicity induced in mongrel dogs [29,30,53]. Amrinone temporarily increased cardiac output, but had no effect on heart rate [30]. Milrinone increased cardiac output in mongrel dogs compared to controls but had no effect on heart rate [29,53]. Measures of survival between groups were not reported.

Wistar rats who had received an infusion of intravenous propranolol 24 mg/kg/h targeting a MAP of 50% of baseline resulted had significantly longer survival times compared to controls when administered either 10% fructose disphosphate 125 mg/kg or 250 mg/kg following by an infusion of 20 mg/kg/h (140 min for both interventions versus 80 and 90 min for glucose controls) [47]. Animals treated with fructose disphosphate 250 mg/kg also showed a statistically significant increase in mean and systolic blood pressure [47].

One study examined the effect of levosimendan (either 36 µg/kg loading dose alone or following by an infusion of 0.6 µg/kg/min) administered to Wistar rats following commencement of an infusion of metoprolol (24 mg/kg/h) led to a statistically significant increase in cardiac output and survival time compared to controls and adrenaline (0.5 µm/kg/min) [51]. A study by the same authors of Wistar rats administered levosimendan following commencement of an infusion of 24 mg/kg/h intravenous propranolol targeting a MAP of 50% of baseline had lower systolic blood pressures compared to controls and no difference in heart rate, QRS interval duration, cardiac output or survival times irrespective of dose [50].

Results: mechanical interventions

Dialysis

Evidence for the use of haemodialysis for the treatment of beta-blocker toxicity is limited to four human case reports. Atenolol intoxication treated with continuous veno-veno haemodiafiltration (CVVHDF) in two patients with anuric renal failure cleared atenolol at a rate comparable to normal renal function (atenolol plasma half-life approximately 24 h) [86,138].

Intermittent haemodialysis utilised in a patient with renal impairment demonstrated an atenolol plasma half-life of 5 h [139]. In addition, haemodialysis used with massive sotalol overdose decreased the apparent half-life of the drug (4.5 h during versus 41 h post haemodialysis) with an extraction ratio of 29.4% [140]. Charcoal haemoperfusion utilised in a patient with a mixed diltiazem and metoprolol overdose resulted in a calculated metoprolol clearance rate of 100 mL/min [141].

Veno-arterial extracorporeal membrane oxygenation (V-A ECMO)

There was one observational study, four case series and 11 case reports pertaining to V-A ECMO. In the event of shock refractory to pharmacotherapy in beta-blocker overdose, V-A ECMO was reported as beneficial in improving haemodynamics in four case series [35,39,62,142] and the comparative observational study [31]. Despite this, mortality rates in these studies are variable. Eight case reports [80,81,112,143–147] reported survival.

A reported side effect from these cases has been limb ischemia associated with the vascular lines inserted [31,39]. This complication, however, is dependent on both expertise and technique. Furthermore, newer catheters and close monitoring of limb perfusion may reduce the risk of ECMO-associated limb ischaemia. Typically, numerous agents have been used prior to the institution of V-A ECMO.

Pacing and other devices

There was one case series and one case report examining pacing in the setting of beta-blocker overdose. The use of temporary overdrive cardiac pacing may result in an increase in heart rate and prevention of arrhythmias, such as torsades des pointes in a case series of four patients who all survived following sotalol toxicity [54]. Two patients had known ingestion amounts (3.6 g and 16 g) and pacing was commenced within 4 h for three cases and the last patient was paced following the failure of lignocaine to abort multiple short runs of ventricular tachycardia [54]. A case report also demonstrated utility in improving haemodynamics once instituted [127].

Two case reports showed the successful utility of an intra-aortic balloon pump to augment blood pressure in patients with refractory hypotension caused by beta-blocker toxicity; however, there was no control arm [148,149]. The findings were contrary to the findings of Thiele et al. [150] which did

not show any benefit in cardiogenic shock, albeit a different patient population.

Results: synthesis

The quality of evidence examining beta-blocker poisoning was either low or very low as per the GRADE methodology [8] (Appendix 5). There was often confounding due to the use of multiple treatments as well as absence of detail, in particular, regarding time of administration and reported effect.

Risk of bias across studies

The overall scoring of studies using the tools previously described (see [Supplementary Materials](#) and [Appendix 5](#)), in general, revealed a considerably high risk of bias such as publication bias, confirmation bias, confounding, and observer bias.

Virtually all case series included in the review were missing at least half the required elements recommended by the Institute of Health Economics tool for quality of case series and quality of reporting. Most studies scored only 10/20 or less, with one case series pertaining to pacing scoring highest at 11/20 [54].

Similarly, included animal studies also performed poorly when assessed using the NRCNA checklist, scoring 9/16 or less. However, when assessed using the ARRIVE checklist, the scores were commonly greater than 10/20.

Insufficient information pertaining to the conditions of laboratory animals, type of species studied, analysis undertaken (e.g., use of blood concentrations, necropsy) and lack of randomisation or concealment is thought to have contributed to the difference between the two checklists.

The risk of bias was high with all studies for each intervention. As expected, the risk of publication bias was estimated to be high with case reports and was all of very low quality.

Mortality

Catecholamines, vasopressors, high-dose insulin euglycaemic therapy and ECMO were associated with reduction in mortality [11–13,31], although it must be acknowledged that multiple treatments were often given simultaneously and details regarding timing of effect compared to timing administration were lacking in some studies. Human studies did not demonstrate an obvious survival benefit associated with utilisation of glucagon or lipid emulsion therapy [21,28,55,68,85]. Similarly, case reports pertaining to atropine and calcium were of very low quality and did not demonstrate an appreciable mortality benefit.

Furthermore, mortality in patients treated with high-dose insulin euglycaemic therapy after beta-blocker poisoning appeared to be similar to that in patients with calcium channel blocker or calcium channel blocker + beta-blocker poisoning. However, this was only noted in two small case series [38,55].

The evidence for mortality benefit from the administration of glucagon [77–79,97–102] was confounded by quality of reporting. While glucagon commonly demonstrated a chronotropic effect at high doses, this effect may be more limited in severe poisoning. Evidence for an effect from glucagon on blood pressure is far less compelling. In most case reports and case series, the effect of glucagon was confounded by the simultaneous use of multiple inotropic and pressor agents. Furthermore, accessibility to these high doses of glucagon may be limited by lack of supply in most hospitals. At least, 5–10 mg as a loading-dose and similar doses are required by infusion per hour to maintain a significant chronotropic effect.

In regards to dialysis in humans, case reports demonstrate that drug clearance is increased for a number of beta-blockers, although there is insufficient evidence to conclude that dialysis changed clinical outcome, in particular, mortality [86,138,139]. Although it is difficult to distinguish whether dialysis changed survival outcome from the limited data, it might be considered for massive overdose of specific beta-blockers (e.g., atenolol or sotalol).

There was a reduction in mortality with the use of ECMO in patients with beta-blocker poisoning and cardiac arrest or severe shock refractory to standard advanced life support measures [80,81,112,143–147]. Overall, V-A ECMO should be considered in patients with refractory shock not responding to other non-invasive therapies.

Haemodynamics

In human studies, haemodynamic improvements in blood pressure and cardiac output were seen with the use of catecholamines, vasopressin and high-dose insulin euglycaemic therapy [11–13]. There were insufficient data in these studies to determine these agents' effect on heart rate, although vasopressin and high-dose insulin euglycaemic therapy are not chronotropic agents.

There were mixed results for high-dose insulin euglycaemic therapy with regards to improvement in haemodynamics [25,43,57]. High-dose insulin euglycaemic therapy appears more effective when used in combination with other interventions such as catecholamines.

There may be a mild to moderate clinical chronotropic benefit with the use of glucagon, but available supply and maintaining therapy is problematic and reduces the practical benefit of glucagon in the clinical setting. Both atropine and glucagon corrected bradycardia, but did not confer any major benefits to mean arterial blood pressure in human [22,65] or animal studies [29,30,36,53].

In addition, the improved haemodynamics with administration of methylthioninium chloride [104–107] in beta-blocker toxicity was confounded by co-ingestion with a calcium channel blocker.

Pacing improved haemodynamics in one case series by improving heart rate with no further deterioration in the patients observed [54]. In cases of left ventricular dysfunction, left ventricular assist devices may also be useful to

augment cardiac output until beta-blocker toxicity abates [86], but there is a lack of controlled studies.

VA-ECMO improved haemodynamics in five human studies [31,35,39,62,142]. It should be noted that very few studies included haemodynamic improvement as an outcome of interest, limiting the ability to recommend one treatment over another.

Animal studies [26,37,58] reported an improvement in haemodynamic measures (cardiac index, maximal ventricular pressure change over time, stroke volume, blood pressure and cardiac conduction) with the use of calcium. However, there was a mixed response to improvement in haemodynamics in the human case reports.

In other animal studies, amrinone, milrinone and fructose di-phosphate led to some improve in cardiac output [29,30,47]. Levosimendan improved blood pressure in one animal study but worsened it in another [50,51].

Effect of treatments on other outcome variables

There was only one study that assessed length of stay as an outcome measure. The authors found that there was no difference in ICU or hospital length of stay for patients with beta-blocker toxicity treated with either catecholamines or high-dose insulin euglycaemic therapy [33].

A few studies reported on the occurrence or frequency of adverse effects resulting from treatments used in beta-blocker poisoning [12,31,39,41,61,95,125]. Adverse events were best described for high-dose insulin euglycaemic therapy, with hypoglycaemia and hypokalaemia the most common effects reported.

Another case series of high-dose insulin euglycaemic therapy use reported that increasing insulin dose did not correlate with the increasing severity of hypoglycaemia or hypokalaemia. It also did not predict the amount of glucose or potassium supplementation required [61]. Adverse effects to high-dose glucagon are also better documented than for many other treatments. Nausea and vomiting were described commonly in two studies [79,98].

Adverse events with ECMO use are well documented. Limb ischemia is the most commonly reported adverse effect. Intravenous lipid emulsion therapy can result in lipaemic blood which may occlude haemodialysis filters and ECMO oxygen exchange circuits as well as interfere with laboratory assay analysis.

A suggested treatment approach for a patient with beta-blocker toxicity

Although the grade of evidence was low to very low, a suggested treatment order for beta-blocker poisoning based on the findings of this review is shown in [Box 1](#).

Whilst a common initial therapy for shock, intravenous fluids should be utilised judiciously as shock secondary to poisoning from beta-blocker is cardiogenic and not hypovolemic in nature. Excess fluid administration may complicate recovery through inadvertent pulmonary oedema. Decontamination with activated charcoal should be

Box 1. Order treatment guide based on Review Findings for beta-blocker toxicity. Each intervention reviewed is given a GRADE [8] classification based upon quality of evidence.

Suggested management for the treatment of shock secondary to beta-blocker poisoning

- Judicious use of intravenous (IV) fluids (10–20 ml/kg crystalloid) to maintain euvoemia.
- Consider decontamination with activated charcoal in secured airway as per current guidelines* (Grade: VERY LOW).
- If bradycardic, this may be response to atropine (Grade: VERY LOW).
- If no improvement, commence norepinephrine or epinephrine (or other inotrope) and titrate (Grade: LOW). The latter may be beneficial in rate-related hypotension where an increase in heart rate would be advantageous.
- If no improvement, assess haemodynamics using a bedside ultrasound to guide next treatment.
 - If hyperdynamic circulation (e.g. no evidence of cardiac pump failure), consider IV vasopressin and increasing vasopressors (Grade: LOW).
 - If reduced contractility, consider adding high dose insulin euglycaemic therapy (HIET) to treatment (Grade: LOW).
- If refractory hypotension despite chemical intervention, consider mechanical extracorporeal life support if available (Grade: LOW).

*Timing of administration will be dependent on different preparations (i.e., modified release) and may change based on collaborative workgroup recommendations.

considered in a patient with a protected airway. Catecholamines such as norepinephrine and epinephrine may be used to treat vasoplegia or cardiogenic shock. A recommendation for which catecholamine to use, either alone or in combination to treat severe beta-blocker toxicity cannot be established from the current evidence.

Bedside cardiac ultrasound can be helpful to distinguish between vasoplegia (for which another agent such as vasopressin would be an adjunctive treatment) and cardiogenic shock, where high-dose insulin euglycaemic therapy may be appropriate in addition to other inotropes. Clinicians should be cognisant of high-dose insulin-associated hypoglycaemia and hypokalaemia, which can persist following cessation of insulin therapy despite the dose administered [61]. In instances where cardiogenic shock is refractory to the above therapies, early consideration of ECMO, if available, may be lifesaving.

Special consideration should be given to propranolol and sotalol given their additional pharmacodynamic effects as described in Table 1. Seizures secondary to propranolol should be managed with titrated IV benzodiazepines. With regards to sotalol, the management of QT prolongation and torsades des pointes is well-described elsewhere [151].

Limitations

The evidence for the management of beta-blocker poisoning was derived from a heterogeneous array of literature prone to various forms of bias, and in almost all cases a low grade of evidence. Different analytical tools were used to assess each type of study. This is not the first time these tools have been used in assessment of toxicology treatments, with the most recent example being a systematic review of treatments for the management of calcium channel blocker toxicity [9].

All clinical and most animal studies were not controlled comparisons of treatments. Most research studies lacked a placebo control arm. It is more than likely that significant publication bias exists due to the more frequent reporting of

positive outcomes to treatment in clinical cases and case series.

Conclusions

Overall, the evidence for the effectiveness of treatments in the management of beta-blocker toxicity is of low quality and drawn predominantly from case series, animal studies and case reports. The majority of this evidence is potentially subject to bias. There appears to be some mortality benefit with the use of catecholamines, high-dose insulin euglycaemic therapy and V-A ECMO. There is insufficient reporting of haemodynamic benefit (beyond case reports of varying quality) to recommend any treatment over another. Notably, glucagon appears to have a minimal effect on improving haemodynamics.

Although evidence is lacking to provide specific recommendations, it is reasonable to have a graduated response to cardiovascular instability beginning with judicious use of IV fluids then choice of single or combination catecholamine/inotrope/vasopressor depending on the haemodynamic compromise evident (bradycardia, left ventricular dysfunction, vasodilatation), addition of high-dose insulin euglycaemic therapy as an adjunctive inotrope and lastly more invasive methods such as V-A ECMO.

Outcomes pertaining to length of stay were seldom reported; however, adverse effects were noted most frequently with insulin causing metabolic derangement. To truly address the appropriate management of beta-blocker toxicity, well-designed, prospective studies are required to evaluate therapies without the burden of confounding variables.

Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

Joe-Anthony Rotella  <http://orcid.org/0000-0003-2068-8757>
 Helen Baxter  <http://orcid.org/0000-0001-6579-8584>

References

- [1] MIMS Online. In Cardiovascular system – beta-adrenergic blocking agents. Crows Nest, NSW: MIMS Australia Pty Ltd.
- [2] Tamargo J, Le Heuzey JY, Mabo P. Narrow therapeutic index drugs: a clinical pharmacological consideration to flecainide. *Eur J Clin Pharmacol*. 2015;71(5):549–567.
- [3] Lauterbach M. Clinical toxicology of beta-blocker overdose in adults. *Basic Clin Pharmacol Toxicol*. 2019;125(2):178–186.
- [4] Gummin DD, Mowry JB, Spyker DA, et al. 2018 Annual Report of the American Association of Poison Control Centers' National Poison Data System (NPDS): 36th Annual Report. *Clin Toxicol (Phila)*. 2019;57:1–194.
- [5] Frishman W, Jacob H, Eisenberg E, et al. Clinical pharmacology of the new beta-adrenergic blocking drugs. Part 8. Self-poisoning with beta-adrenoceptor blocking agents: recognition and management. *Am Heart J*. 1979;98(6):798–811.
- [6] Frishman WH, Alwarshetty M. Beta-adrenergic blockers in systemic hypertension: pharmacokinetic considerations related to the current guidelines. *Clin Pharmacokinet*. 2002;41(7):505–516.
- [7] Brouwers MC, for the AGREE Next Steps Consortium, Kho ME, Browman GP, et al. AGREE II: advancing guideline development, reporting and evaluation in health care. *CMAJ*. 2010;182(18):E839–E842.
- [8] Terracciano L, Brozek J, Compalati E, et al. GRADE system: new paradigm. *Curr Opin Allergy Clin Immunol*. 2010;10(4):377–383.
- [9] St-Onge M, Dubé P-A, Gosselin S, et al. Treatment for calcium channel blocker poisoning: a systematic review. *Clin Toxicol (Phila)*. 2014;52(9):926–944.
- [10] Lavergne V, Nolin TD, Hoffman RS, et al. The EXTRIP (EXtracorporeal TReatments In Poisoning) workgroup: guideline methodology. *Clin Toxicol*. 2012;50(5):403–413.
- [11] Cox D, et al. A comparison of vasopressor utility for drug overdose-induced shock. *Clin Toxicol*. 2015;53(4):296.
- [12] Robinson HP, et al. Combined therapy with high-dose insulin and vasoactive drugs for beta blocker/calcium channel blocker overdose: characteristics, complications and outcomes. *Clin Toxicol*. 2015;53(7):651.
- [13] Musselman M, Farber M, Smolinske S, et al. A comparison of high dose insulin therapy and conventional inotropic therapy in calcium channel blocker and/or beta-blocker toxic ingestions. *Ann Emerg Med*. 2011;58(4):S325.
- [14] von Elm E, Altman DG, Egger M, et al. The Strengthening of Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. *J Clin Epidemiol*. 2008;61(4):344–349.
- [15] Institute of Medicine (US) and National Research Council (US) Committee on the Framework for Evaluating the Safety of Dietary Supplements. Dietary supplements: a framework for evaluating safety. Washington (DC): National Academies Press; 2005.
- [16] Moga CGB, Schopflocher D, Harstall C. Development of a Quality Appraisal Tool for case series studies using a modified Delphi technique. Edmonton AB: Institute of Health Economics; 2012.
- [17] Kilkenny C, Browne WJ, Cuthill IC, et al. Improving bioscience research reporting: the ARRIVE guidelines for reporting animal research. *PLoS Biol*. 2010;8(6):e1000412.
- [18] National Research Council. Guidance for the description of animal research in scientific publications. Washington (DC): The National Academies Press; 2011.
- [19] Ryan W, Wong A, Graudins A. Analysis of beta blocker and calcium channel blocker overdoses treated with high-dose insulin euglycaemic therapy and/or catecholamine infusion. *J Med Toxicol*. 2018;14(1):50–51.
- [20] Jovic-Stosic J, et al. Antidotal use of intravenous lipid emulsion: 5 years' experience in an intensive care unit. *Clin Toxicol*. 2016; 54(4):476.
- [21] Cole JB, Stellpflug SJ, Engebretsen KM. Asystole immediately following intravenous fat emulsion for overdose. *J Med Toxicol*. 2014;10(3):307–310.
- [22] Kandil A, Abd-Elatif M, Abd-Elwahed K. Aversion of the cardiac toxicity of propranolol by atropine. *J Drug Res*. 1979;11(1–2): 17–26.
- [23] Tabone D. BET 2: intralipid/lipid emulsion in beta-blocker overdose. *Emerg Med J*. 2011;28(11):991–993.
- [24] Meehan TJ, et al. Beta blocker toxicity successfully treated with intravenous fat emulsion: a case series. *Clin Toxicol*. 2009;47(7): 735.
- [25] Cole JB, Stellpflug SJ, Ellsworth H, et al. A blinded, randomized, controlled trial of three doses of high-dose insulin in poison-induced cardiogenic shock. *Clin Toxicol*. 2013;51(4):201–207.
- [26] Langemeijer J, de Wildt D, de Groot G, et al. Calcium interferes with the cardiodepressive effects of beta-blockers overdose in isolated rat hearts. *J Toxicol – Clin Toxicol*. 1986;24(2):111–133.
- [27] Jacques AC. Case series of hyperinsulinaemic euglycaemic therapy and lipid emulsion therapy in self-poisoning with calcium channel antagonists or beta-blockers. *J Intensive Care Soc*. 2014; 15(1 suppl):S20–S21.
- [28] Mithani S, Dong K, Wilmott A, et al. A cohort study of unstable overdose patients treated with intravenous lipid emulsion therapy. *CJEM Can J Emerg Med Care*. 2017;19(04):256–264.
- [29] Sato S, Tsuji MH, Okubo N, et al. Combined use of glucagon and milrinone may not be preferable for severe propranolol poisoning in the canine model. *J Toxicol – Clin Toxicol*. 1995;33(4): 337–342.
- [30] Love JN, Leasure JA, Mundt DJ, et al. A comparison of amrinone and glucagon therapy for cardiovascular depression associated with propranolol toxicity in a canine model. *J Toxicol – Clin Toxicol*. 1992;30(3):399–412.
- [31] Masson R, Colas V, Parienti J-J, et al. A comparison of survival with and without extracorporeal life support treatment for severe poisoning due to drug intoxication. *Resuscitation*. 2012; 83(11):1413–1417.
- [32] Holger JS, Engebretsen KM, Obetz CL, et al. A comparison of vasopressin and glucagon in beta-blocker induced toxicity. *Clin Toxicol: Off J Am Acad Clin Toxicol Eur Assoc Poisons Centres Clin Toxicologists*. 2006;44(1):45–51.
- [33] Rials SJ, Tse WW. Effects of atropine on cardiac arrest induced by propranolol and digitoxin. *Federation Proc*. 1979;38 (3 II): 4695.
- [34] Megarbane B, Leprince P, Deye N, et al. Emergency feasibility in medical intensive care unit of extracorporeal life support for refractory cardiac arrest. *Intensive Care Med*. 2007;33(5):758–764.
- [35] Daubin C, Lehoux P, Ivascau C, et al. Extracorporeal life support in severe drug intoxication: a retrospective cohort study of seventeen cases. *Crit Care*. 2009;13(4):R138.
- [36] Kosinski EJ, Malindzak GS Jr. Glucagon and isoproterenol in reversing propranolol toxicity. *Arch Intern Med*. 1973;132(6): 840–843.
- [37] Love JN, Hanfling D, Howell JM. Hemodynamic effects of calcium chloride in a canine model of acute propranolol intoxication. *Ann Emerg Med*. 1996;28(1):1–6.
- [38] Cole JB, Arens AM, Laes JR, et al. High dose insulin for beta-blocker and calcium channel-blocker poisoning: 17 years of experience from a single poison center. *Am J Emerg Med*. 2018; 06:06.
- [39] Pozzi M, Koffel C, Djaref C, et al. High rate of arterial complications in patients supported with extracorporeal life support for drug intoxication-induced refractory cardiogenic shock or cardiac arrest. *J Thorac Dis*. 2017;9(7):1988–1996.
- [40] Doecker B, Healy W, Cortez E, et al. High-dose insulin and intravenous lipid emulsion therapy for cardiogenic shock induced by

- intentional calcium-channel blocker and Beta-blocker overdose: a case series. *J Emerg Med.* 2014;46(4):486–490.
- [41] Holger JS, Stellpflug SJ, Cole JB, et al. High-dose insulin: a consecutive case series in toxin-induced cardiogenic shock. *Clin Toxicol: Off J Am Acad Clin Toxicol Eur Assoc Poisons Centres Clin Toxicologists.* 2011;49(7):653–658.
 - [42] Kerns W, Schroeder D, Williams C 2nd, et al. Insulin improves survival in a canine model of acute beta-blocker toxicity. *Ann Emerg Med.* 1997;29(6):748–757.
 - [43] Harvey M, Cave G, Lahner D, et al. Insulin versus lipid emulsion in a rabbit model of severe propranolol toxicity: a pilot study. *Crit Care Res Pract.* 2011;2011:1–7.
 - [44] Holger JS, Engebretsen KM, Fritzlir SJ, et al. Insulin versus vasopressin and epinephrine to treat beta-blocker toxicity. *Clin Toxicol: Off J Am Acad Clin Toxicol Eur Assoc Poisons Centres Clin Toxicologists.* 2007;45(4):396–401.
 - [45] Eren Cevik S, Tasyurek T, Guneysele O. Intralipid emulsion treatment as an antidote in lipophilic drug intoxications. *Am J Emerg Med.* 2014;32(9):1103–1108.
 - [46] Harvey MG, Cave GR. Intralipid infusion ameliorates propranolol-induced hypotension in rabbits. *J Med Toxicol.* 2008;4(2):71–76.
 - [47] Kalam Y, Graudins A. Intravenous fructose-1,6-diphosphate improves survival and blood pressure in propranolol-poisoned rodents. *Clin Toxicol.* 2012;50(7):546–554.
 - [48] Browne A, Harvey M, Cave G. Intravenous lipid emulsion does not augment blood pressure recovery in a rabbit model of metoprolol toxicity. *J Med Toxicol.* 2010;6(4):373–378.
 - [49] Gunaydin YK. Intravenous lipid emulsion therapy in calcium channel blocker and beta-blocker poisoning. *Notfall Und Rettungsmedizin.* 2017;20(1):54–60.
 - [50] Kalam Y, Graudins A. Levosimendan does not improve cardiac output or blood pressure in a rodent model of propranolol toxicity when administered using various dosing regimens. *Int J Toxicol.* 2012;31(2):166–174.
 - [51] Kalam Y, Graudins A. Levosimendan infusion improves cardiac output but not blood pressure in a rodent model of severe metoprolol toxicity. *Hum Exp Toxicol.* 2012;31(9):955–963.
 - [52] Cave G, Harvey M. Lipid emulsion may augment early blood pressure recovery in a rabbit model of atenolol toxicity. *J Med Toxicol.* 2009;5(1):50–51.
 - [53] Sato S, Tsuji MH, Okubo N, et al. Milrinone versus glucagon: comparative hemodynamic effects in canine propranolol poisoning. *J Toxicol – Clin Toxicol.* 1994;32(3):277–289.
 - [54] Totterman KJ, Turto H, Pellinen T. Overdrive pacing as treatment of sotalol induced ventricular tachyarrhythmias (torsade de pointes). *Acta Med Scand.* 2009;212(S668):28–33.
 - [55] Gerwin H, Behrman A, Goetz R, et al. Predictors of response to insulin, glucagon, and lipid emulsion in antihypertensive overdose. *Crit Care Med.* 2015;43(1):32.
 - [56] Andersson T, Heath A, Mattsson H. Prenalator as an antidote to massive doses of metoprolol – a cardiovascular study in the dog. *Acta Med Scand.* 2009;211(S659):71–88.
 - [57] Katzung KG, Leroy JM, Boley SP, et al. A randomized controlled study comparing high-dose insulin to vasopressors or combination therapy in a porcine model of refractory propranolol-induced cardiogenic shock. *Clin Toxicol (Phila).* 2019;57:1–7.
 - [58] Whitehurst VE, Joseph X, Vick JA, et al. Reversal of propranolol and verapamil toxicity by calcium. *Vet Hum Toxicol.* 1996;110(1–3):110–113.
 - [59] Cave G, Harvey MG, Castle CD. The role of fat emulsion therapy in a rodent model of propranolol toxicity: a preliminary study. *J Med Toxicol.* 2006;2(1):4–7.
 - [60] Sebe A, Dişel NR, Açıkalın Akpınar A, et al. Role of intravenous lipid emulsions in the management of calcium channel blocker and beta-blocker overdose: 3 years experience of a university hospital. *Postgrad Med.* 2015;127(2):119–124.
 - [61] Page CB, Ryan NM, Isbister GK. The safety of high-dose insulin euglycaemia therapy in toxin-induced cardiac toxicity. *Clin Toxicol: Off J Am Acad Clin Toxicol Eur Assoc Poisons Centres Clin Toxicologists.* 2018;56(6):389–396.
 - [62] Babatasi G, Massetti M, Verrier V, et al. Severe intoxication with cardiotoxic drugs: value of emergency percutaneous cardiocirculatory assistance. *Arch Mal Coeur Vaiss.* 2001;94(12):1386–1392.
 - [63] Riggan M, McLeod SL, Hames H. The use of intralipid and high dose insulin therapy in the emergency department management of beta blocker and calcium channel blocker toxicity. *Can J Emerg Med.* 2014;16 (Supp 1):S52.
 - [64] Darracq MA, Thornton SL, Do HM, et al. Utilization of hyperinsulinemia euglycemia and intravenous fat emulsion following poison center recommendations. *J Med Toxicol.* 2013;9(3):226–230.
 - [65] Catlin J, et al. Weight-based dosing for intravenous glucagon bolus is associated with chronotropic response in beta-blocker toxicity. *J Med Toxicol.* 2018;14(1):53.
 - [66] Smolinske S, Hoffman RS, Villeneuve E, et al. Observational study on the use of methylene blue for shock in fatal overdoses. *Clin Toxicol.* 2019;57(3):197–202.
 - [67] Harley C, Kannan S, Morrow B. Combined overdose with bisoprolol, nifedipine and diltiazem. *Care Crit Ill.* 2001;17(2):65–66.
 - [68] Abbasi IA, Sorsby S. Prolonged toxicity from atenolol overdose in an adolescent. *Clin Pharm.* 1986;5(10):836–837.
 - [69] Amundson DE, Brodine SK. A fatal case of propranolol poisoning. *Drug Intell Clin Pharm.* 1988;22(10):781–782.
 - [70] Henry M, Kay MM, Vercellio P. Cardiogenic shock associated with calcium-channel and beta blockers: reversal with intravenous calcium chloride. *Am J Emerg Med.* 1985;3(4):334–336.
 - [71] Pertoldi F, D'Orlando L, Mercante WP. Electromechanical dissociation 48 hours after atenolol overdose: usefulness of calcium chloride. *Ann Emerg Med.* 1998;31(6):777–781.
 - [72] Barton CA, Johnson NB, Mah ND, et al. Successful treatment of a massive metoprolol overdose using intravenous lipid emulsion and hyperinsulinemia/euglycemia therapy. *Pharmacotherapy.* 2015;35(5):e56–e60.
 - [73] Czerwonka E, Heim M. Lipid rescue therapy and hyperinsulinemia/euglycemia for atenolol and zolpidem overdose. *Crit Care Med.* 2015;43 (12 (supplement 1):316.
 - [74] Markota A, Hajdinjak E, Rupnik B, et al. Treatment of near-fatal beta blocker and calcium channel blocker intoxication with hyperinsulinemic euglycemia, intravenous lipid emulsions and high doses of norepinephrine. *Signa Vitae.* 2015;10(1):144–150.
 - [75] Qureshi M, Catalasan G. The use of high dose insulin euglycemia therapy in calcium channel blocker and beta blocker overdose. *J Gen Intern Med.* 2016;31(2 Supplement):S760–S761.
 - [76] Seegobin K, Maharaj S, Deosaran A, et al. Severe beta blocker and calcium channel blocker overdose: role of high dose insulin. *Am J Emerg Med.* 2018;36(4):736.e5–736.e6.
 - [77] Ray S, Garg A, Singh VK, et al. Severe suicidal digoxin and propranolol toxicity with insulin overdose: a case report. *Indian J Crit Care Med.* 2014;18(3):173–175.
 - [78] Bouchard NC, Forde J, Hoffman RS. Carvedilol overdose with quantitative confirmation. *Basic Clin Pharmacol Toxicol.* 2008;103(1):102–103.
 - [79] Fung HT, Lai CH, Wong OF, et al. The use of glucagon and other antidotes in a case of beta-blocker and calcium channel blocker overdose. *Hong Kong J Emerg Med.* 2007;14(2):113–118.
 - [80] Kacirowa I, Grundmann M, Kolek M, et al. Lethal suicide attempt with a mixed-drug intoxication of metoprolol and propafenone – a first pediatric case report. *Forensic Sci Int.* 2017;278:e34–e40.
 - [81] Kolcz J, Pietrzyk J, Januszewska K, et al. Extracorporeal life support in severe propranolol and verapamil intoxication. *J Intensive Care Med.* 2007;22(6):381–385.
 - [82] Murphy NG, et al. Early use of high dose insulin-glucose euglycemia prevents hemodynamic instability after metoprolol overdose. *Clin Toxicol.* 2010;48(6):648.
 - [83] Paetow GM, et al. Use of tissue perfusion monitoring to titrate high dose insulin in beta blocker and calcium channel blocker overdose. *Clin Toxicol.* 2015;53(7):705–706.
 - [84] Tai YT, Lo CW, Chow WH, et al. Successful resuscitation and survival following massive overdose of metoprolol. *Br J Clin Pract.* 1990;44(12):746–747.

- [85] Salzberg MR, Gallagher EJ. Propranolol overdose. *Ann Emerg Med.* 1980;9(1):26–27.
- [86] Heise CW, Beutler D, Bosak A, et al. Massive atenolol, lisinopril, and chlorthalidone overdose treated with endoscopic decontamination, hemodialysis, impella percutaneous left ventricular assist device, and ECMO. *J Med Toxicol.* 2015;11(1):110–114.
- [87] Jones JW, Clark MA, Mullen BL. Suicide by ingestion of propranolol. *J Forensic Sci.* 1982;27(1):213–216.
- [88] Bourg PW, McLarnon A. Propranolol overdose. *J Emerg Nurs.* 1982;8(6):275–276.
- [89] Bourguin J, et al. Acute poisoning with sustained release propranolol and acetaminophen: delayed effects. *Fundam Clin Pharmacol.* 2014;28 (suppl 1):98.
- [90] Halloran TJ, Phillips CE. Propranolol intoxication. A severe case responding to norepinephrine therapy. *Arch Intern Med.* 1981; 141(6):810–811.
- [91] Hopkins LE, Sunkersing J, Jacques A. Too many pills to swallow: a case of a mixed overdose. *J Intensive Care Soc.* 2017;18(3): 247–250.
- [92] Lewis M, Kallenbach J, Germond C, et al. Survival following massive overdose of adrenergic blocking agents (acebutolol and labetalol). *Eur Heart J.* 1983;4(5):328–332.
- [93] Jones JL. Metoprolol overdose. *Ann Emerg Med.* 1982;11(2): 114–115.
- [94] O'Grady J, Anderson S, Pringle D. Successful treatment of severe atenolol overdose with calcium chloride. *CJEM Can J Emerg Med Care.* 2001;3(03):224–227.
- [95] Cole JB, et al. High dose insulin for beta blocker and calcium channel blocker poisoning: assessment of feasibility and metabolic abnormalities from 17 years of experience from a single poison center. *Acad Emerg Med.* 2018;25 (Supplement 1): S216–S217.
- [96] Page C, Hacket LP, Isbister GK. The use of high-dose insulin-glucose euglycemia in beta-blocker overdose: a case report. *J Med Toxicol.* 2009;5(3):139–143.
- [97] Love JN, Elshami J. Cardiovascular depression resulting from atenolol intoxication. *Eur J Emerg Med.* 2002;9(2):111–114.
- [98] DeLima LG, Kharasch ED, Butler S. Successful pharmacologic treatment of massive atenolol overdose: sequential hemodynamics and plasma atenolol concentrations. *Anesthesiology.* 1995;83(1):204–207.
- [99] Mansell PI. Glucagon in the management of deliberate self-poisoning with propranolol. *Arch Emerg Med.* 1990;7(3):238–240.
- [100] O'Mahony D, O'Leary P, Molloy MG. Severe oxprenolol poisoning: the importance of glucagon infusion. *Hum Exp Toxicol.* 1990;9(2):101–103.
- [101] Khan MI, Miller MT. Beta-blocker toxicity—the role of glucagon. Report of 2 cases.[Erratum appears in *S Afr Med J* 1986 Jul 19;70(2):124]. *S Afr Med J. Suid-Afrikaanse Tydskrif Vir Geneeskunde.* 1985;67(26):1062–1063.
- [102] Ward DE, Jones B. Glucagon and beta-blocker toxicity. *Br Med J.* 1976;2(6028):151.
- [103] Khan MI, Miller MT. Beta-blocker toxicity –the role of glucagon. Report of 2 cases. *S Afr Med J.* 1985;67(26):1062–1063.
- [104] Aggarwal N, Kupfer Y, Seneviratne C, et al. Methylene blue reverses recalcitrant shock in beta-blocker and calcium channel blocker overdose. *BMJ Case Rep.* 2013;2013 (1):bcr2012007402.
- [105] Kim HK, et al. Methylene blue for refractory hypotension in cardiovascular drug overdose. *Clin Toxicol.* 2012;50(4):365.
- [106] Laes JR, et al. Hemodynamic changes after methylene blue administration in poison-induced shock. *Clin Toxicol.* 2014;52(7): 753.
- [107] Laes JR, Williams DM, Cole JB. Improvement in hemodynamics after methylene blue administration in drug-induced vasodilatory shock: a case report. *J Med Toxicol.* 2015;11(4):460–463.
- [108] Biary R, et al. Intravenous lipid emulsion used in the therapy of a patient with prolonged cardiac pauses following a single pill ingestion of propafenone. *Clin Toxicol.* 2014;52(4):405.
- [109] Carr D, et al. Successful resuscitation of a carvedilol overdose using intravenous fat emulsion (IFE). *Clin Toxicol.* 2009;47(7):727.
- [110] Cumpston KL, Gopaul R, Wills B. Adjunctive use of low-dose intralipid associated with hemodynamic improvement in combined amlodipine and labetalol overdose refractory to standard therapy. *Am J Ther.* 2017;24(4):e485.
- [111] Dean P, Ruddy JP, Marshall S. Intravenous lipid emulsion in propranolol overdose. *Anaesthesia.* 2010;65(11):1148–1150.
- [112] Escajeda JT, Katz KD, Rittenberger JC. Successful treatment of metoprolol-induced cardiac arrest with high-dose insulin, lipid emulsion, and ECMO. *Am J Emerg Med.* 2015;33(8): 1111.e1–4.
- [113] Johnson-Arbor K, Salinger L, Luczycki S. Prolonged laboratory interference after administration of intravenous lipid emulsion therapy. *J Med Toxicol.* 2015;11(2):223–226.
- [114] Jovic-Stosic J, Gligic B, Putic V, et al. Severe propranolol and ethanol overdose with wide complex tachycardia treated with intravenous lipid emulsion: a case report. *Clin Toxicol: Off J Am Acad Clin Toxicol Eur Assoc Poisons Centres Clin Toxicologists.* 2011;49(5):426–430.
- [115] Macala K, Bandrauk N, Smith A. Fatty emulsion: surviving clonidine & propranolol overdose. *Can J Anesthesia.* 2013;60(1):S66.
- [116] Orthober R, Huecker M, Sandlin D, et al. Polypharmacy overdose in a 36-year-old man. *Air Med J.* 2013;32(1):4.
- [117] Pasha A, Kazmi S, Ehtesham M. Is fat always bad? A case study demonstrating the lifesaving effect of lipid emulsion therapy in beta blocker and calcium channel blocker overdose. *Chest.* 2017;152(4):A259.
- [118] Rodriguez B, Wilhelm A, Kokko KE. Lipid emulsion use precluding renal replacement therapy. *J Emerg Med.* 2014;47(6): 635–637.
- [119] Stellpflug SJ, Harris CR, Engebretsen KM, et al. Overdose of diltiazem, metoprolol, and amiodarone treated successfully with intravenous fat emulsion and high dose insulin in an awake patient. *Clin Toxicol.* 2010;48(3):227–229.
- [120] Stellpflug SJ, Fritzlar SJ, Cole JB, et al. Cardiotoxic overdose treated with intravenous fat emulsion and high-dose insulin in the setting of hypertrophic cardiomyopathy. *J Med Toxicol.* 2011;7(2):151–153.
- [121] Troendle MM, et al. Adjunctive use of low dose intralipid associated with hemodynamic improvement in combined amlodipine and labetalol overdose refractory to standard therapy. *Clin Toxicol.* 2013;51(7):650–651.
- [122] Thompson AM, Franco Palacios CR, Henriksen MN. Intravenous lipid emulsion and high-dose insulin as adjunctive therapy for propranolol toxicity in a pediatric patient. *Am J Health-Syst Pharm.* 2016;73(12):880–885.
- [123] Walter E, McKinlay J, Corbett J, et al. Review of management in cardiotoxic overdose and efficacy of delayed intralipid use. *J Intensive Care Soc.* 2018;19(1):50–55.
- [124] Levine M, for the Lipid Emulsion Workgroup, Hoffman RS, Lavergne V, et al. Systematic review of the effect of intravenous lipid emulsion therapy for non-local anesthetics toxicity. *Clin Toxicol.* 2016;54(3):194–221.
- [125] Hayes BD, Lipid Emulsion Workgroup, Gosselin S, Calello DP, et al. Systematic review of clinical adverse events reported after acute intravenous lipid emulsion administration. *Clin Toxicol (Phila).* 2016;54(5):365–404.
- [126] Assimes TL, Malcolm I. Torsade de pointes with sotalol overdose treated successfully with lidocaine. *Can J Cardiol.* 1998;14(5): 753–756.
- [127] Belton P, Sheridan J, Mulcahy R. A case of sotalol poisoning. *Ir J Med Sci.* 1982;151(1):126–127.
- [128] Edvardsson N, Varnauskas E. Clinical course, serum concentrations and elimination rate in a case of massive sotalol intoxication. *Eur Heart J.* 1987;8(5):544–548.
- [129] Sandroni C, Cavallaro F, Caricato A, et al. Enoximone in cardiac arrest caused by propranolol: two case reports. *Acta Anaesthesiol Scand.* 2006;50(6):759–761.
- [130] Sandroni C, Cavallaro F, Addario C, et al. Successful treatment with enoximone for severe poisoning with atenolol and verapamil: a case report. *Acta Anaesthesiol Scand.* 2004;48(6): 790–792.

- [131] Shanker UR, Webb J, Kotze A. Sodium bicarbonate to treat massive beta blocker overdose. *Emerg Med J*. 2003;20(4):393.
- [132] Hoepfer MM, Boeker KH. Overdose of metoprolol treated with enoximone. *N Engl J Med*. 1996;335(20):1538–1539.
- [133] Kollef MH. Labetalol overdose successfully treated with amrinone and alpha-adrenergic receptor agonists. *Chest*. 1994;105(2):626–627.
- [134] Hurwitz MD, Kallenbach JM, Pincus PS. Massive propranolol overdose. *Am J Med*. 1986;81(6):1118.
- [135] Freestone S, Thomas HM, Bhamra RK, et al. Severe atenolol poisoning: treatment with prenalator. *Hum Toxicol*. 1986;5(5):343–345.
- [136] Agura ED, Wexler LF, Witzburg RA. Massive propranolol overdose. Successful treatment with high-dose isoproterenol and glucagon. *Am J Med*. 1986;80(4):755–757.
- [137] Wallin CJ, Hulting J. Massive metoprolol poisoning treated with prenalator. *Acta Med Scand*. 2009;214(3):253–255.
- [138] Pfaender M, Casetti PG, Azzolini M, et al. Successful treatment of a massive atenolol and nifedipine overdose with CVVHDF. *Minerva Anesthesiol*. 2008;74(3):97–100.
- [139] Huang S-HS, Tirona RG, Ross C, et al. Case report: atenolol overdose successfully treated with hemodialysis. *Hemodial Int*. 2013;17(4): n/a–5.
- [140] Singh SN, Lazin A, Cohen A, et al. Sotalol-induced torsades de pointes successfully treated with hemodialysis after failure of conventional therapy. *Am Heart J*. 1991;121(2):601–602.
- [141] Anthony T, Jastremski M, Elliott W, et al. Charcoal hemoperfusion for the treatment of a combined diltiazem and metoprolol overdose. *Ann Emerg Med*. 1986;15(11):1344–1348.
- [142] Megarbane B, Deye N, Baud FJ. Extracorporeal life support for poisonings with cardiotoxicants [French]. *Revue du Praticien*. 2008;58(8):855–860.
- [143] Bilbault P, Pynn S, Mathien C, et al. Near-fatal betaxolol self-poisoning treated with percutaneous extracorporeal life support. *Eur J Emerg Med*. 2007;14(2):120–122.
- [144] Chenoweth JA, Colby DK, Sutter ME, et al. Massive diltiazem and metoprolol overdose rescued with extracorporeal life support. *Am J Emerg Med*. 2017;35(10):1581.e3–1581.e5.
- [145] De Rita F, Barozzi L, Franchi G, et al. Rescue extracorporeal life support for acute verapamil and propranolol toxicity in a neonate. *Artif Organs*. 2011;35(4):416–420.
- [146] Maskell KF, Ferguson NM, Bain J, et al. Survival after cardiac arrest: ECMO rescue therapy after amlodipine and metoprolol overdose. *Cardiovasc Toxicol*. 2017;17(2):223–225.
- [147] Rona R, Cortinovis B, Marcolin R, et al. Extra-corporeal life support for near-fatal multi-drug intoxication: a case report. *J Med Case Rep*. 2011;5(1):231.
- [148] Lane AS, Woodward AC, Goldman MR. Massive propranolol overdose poorly responsive to pharmacologic therapy: use of the intra-aortic balloon pump. *Ann Emerg Med*. 1987;16(12):1381–1383.
- [149] Janion M, Stępień A, Sielski J, et al. Is the intra-aortic balloon pump a method of brain protection during cardiogenic shock after drug intoxication? *J Emerg Med*. 2010;38(2):162–167.
- [150] Thiele H, Zeymer U, Neumann F-J, et al. Intraaortic balloon support for myocardial infarction with cardiogenic shock. *N Engl J Med*. 2012;367(14):1287–1296.
- [151] Thomas SH, Behr ER. Pharmacological treatment of acquired QT prolongation and torsades de pointes. *Br J Clin Pharmacol*. 2016;81(3):420–427.
- [152] Litman RS, Zerangue BA. Cardiac arrest after esmolol administration: a review of acute beta-blocker toxicity. *J Am Osteopath Assoc*. 1996;96(10):616–618.
- [153] Hicks PR, Rankin AP. Massive adrenaline doses in labetalol overdose. *Anaesth Intensive Care*. 1991;19(3):447–449.
- [154] Isbister GK, Ang K, Gorman K, et al. Zero-order metoprolol pharmacokinetics after therapeutic doses: severe toxicity and cardiogenic shock. *Clin Toxicol: Off J Am Acad Clin Toxicol Eur Assoc Poisons Centres Clin Toxicologists*. 2016;54(9):881–885.
- [155] Kalman S, Berg S, Lisander B. Combined overdose with verapamil and atenolol: treatment with high doses adrenergic agonists. *Acta Anaesthesiol Scand*. 1998;42(3):379–382.
- [156] Jones B, Jacobsen D, Helgeland A, et al. Treatment of beta-blocker poisoning. *Lancet*. 1980;315(8176):1031–1032.
- [157] Doepker B, Cortez E, Healy W, et al. High dose insulin and lipid emulsion therapy for cardiogenic shock: a case series. *Crit Care Med*. 2013;41 (12 (supplement 1)):A308–A309.
- [158] Hasin T, Leibowitz D, Antopolsky M, et al. The use of low-dose insulin in cardiogenic shock due to combined overdose of verapamil, enalapril and metoprolol. *Cardiology*. 2006;106(4):233–236.
- [159] Montague AJ, et al. High dose insulin to treat propranolol toxicity in a 7-month-old infant. *Clin Toxicol*. 2015;53(7):740.
- [160] Singh S, Singh S, Chittivelu S. Is using high dose insulin in lieu of glucagon in beta blocker toxicity a good idea? *Chest Conference: CHEST*. 2015;148(4):275A (4 MEETING ABSTRACT).
- [161] Cole JB, et al. Failure of high dose insulin and intravenous fat emulsion in 2 patients with poison-induced cardiogenic shock. *Clin Toxicol*. 2011;49(6):537–538.
- [162] Cole JB, Stelpflug SJ, Ellsworth H, et al. A blinded, randomized, controlled trial of three doses of high-dose insulin in poison-induced cardiogenic shock.[Erratum appears in *Clin Toxicol (Phila)*. 2013 Dec;51(10):1246]. *Clin Toxicol: Off J Am Acad Clin Toxicol Eur Assoc Poisons Centres Clin Toxicologists*. 2013;51(4):201–207.
- [163] Fernandes CM, Daya MR. Sotalol-induced bradycardia reversed by glucagon. *Can Fam Physician*. 1995;41: 659–660, 663.
- [164] Rooney M, Massey KL, Jamali F, et al. Acebutolol overdose treated with hemodialysis and extracorporeal membrane oxygenation. *J Clin Pharmacol*. 1996;36(8):760–763.
- [165] Rygnestad T, Moen S, Wahba A, et al. Severe poisoning with sotalol and verapamil. Recovery after 4 h of normothermic CPR followed by extra corporeal heart lung assist. *Acta Anaesthesiol Scand*. 2005;49(9):1378–1380.
- [166] Nordmark Grass J, Ahlner J, Kugelberg FC, et al. A case of massive metoprolol and amlodipine overdose with blood concentrations and survival following extracorporeal corporeal membrane oxygenation (ECMO). *Clin Toxicol: Off J Am Acad Clin Toxicol Eur Assoc Poisons Centres Clin Toxicologists*. 2019;57(1):66–68.

Appendix 1. Search strategies utilised in beta-blocker systematic review

Ovid Embase (1974–26 November 2019) – 10,395 results

1. beta adrenergic receptor blocking agent/
2. (beta-blocker* or beta antagonist* or beta adrenergic blocker or beta adrenergic receptor block* or beta adrenoreceptor blocker* or beta adrenoreceptor antagonist* or beta-blocking agent* or (block* adj1 beta adrenergic) or prop??nolol or sot?lol or met?p??lol or aten?lol or bis?p??lol or carv?d?lol or neb?v?lol or tim?lol or labet?lol or pind?lol or esm?lol or ox?p??n?lol).tw.
3. 1 or 2
4. intoxication/or drug overdose/or toxicity/
5. (overdose or poisoning or toxicity or intoxication or ingestion* or adverse effect* or medication error*).tw.
6. 4 or 5
7. 3 and 6

Ovid MEDLINE (1946–26 November 2019) – 4266 results

1. Adrenergic beta-Antagonists/
2. (beta-blocker* or beta antagonist* or beta adrenergic blocker or beta adrenergic receptor block* or beta adrenoreceptor blocker* or beta adrenoreceptor antagonist* or beta-blocking agent* or (block* adj1 beta adrenergic) or prop??nolol or sot?lol or

met?p??lol or aten?lol or bis?p??lol or carv?d?lol or neb?v?lol or tim?lol or labet?lol or pind?lol or esm?lol or ox?p??n?lol).tw.

3. 1 or 2

4. POISONING/or Drug Overdose/

5. (overdose or poisoning or toxicity or intoxication or ingestion* or adverse effect* or medication error*).tw.

6. 4 or 5

7. 3 and 6

CENTRAL – Cochrane Central Register of Controlled Trials (as at 26 November 2019) – 6177 results

#1 MeSH descriptor: [Adrenergic beta-Antagonists] this term only

#2 (beta-blocker* or beta antagonist* or beta adrenergic blocker or beta adrenergic receptor block* or beta adrenoreceptor blocker* or beta adrenoreceptor antagonist* or beta-blocking agent*);ti,ab,kw

#3 (block* next beta adrenergic);ti,ab,kw

#4 (prop??nolol or sot?lol or met?p??lol or aten?lol or bis?p??lol or carv?-d?lol or neb?v?lol or tim?lol or labet?lol or pind?lol or esm?lol or ox?p??n?lol);ti,ab,kw

#5 #1 or #2 or #3 or #4

#6 MeSH descriptor: [Poisoning] this term only

#7 MeSH descriptor: [Drug Overdose] this term only

#8 (overdose or poisoning or toxicity or intoxication or ingestion* or adverse effect* or medication error*);ti,ab,kw

#9 #6 or #7 or #8

#10 #5 and #9

Appendix 2. Case reports published focusing on an intervention or reporting a side effect in the setting of beta-blocker overdose

Source	Subject and patient characteristics	Type of BB poisoning	Intervention and co-interventions	Results
Gastric Decontamination				
Harley et al. [67]	Subject: 57-year-old man Co-ingestions: Diltiazem SR 960 mg Nifedipine 120 mg Aspirin 900 mg	Bisoprolol 40 mg	Intervention: Activated charcoal Co-interventions: Adrenaline infusion, haemodiafiltration calcium, glucagon	Mortality: No Morbidity: Acute kidney injury Haemodynamic improvement: NR Secondary outcomes: NR Adverse effects: NR
Abbasi et al. [68]	Subject: 15-year-old girl	Atenolol 500 mg	Intervention: Activated charcoal Co-interventions: Gastric lavage, atropine boluses, glucose infusion	Mortality: No Morbidity: NR Haemodynamic improvement: NR Secondary outcomes: NR Adverse effects: NR
Heise et al. [86]	Subject: 44-year-old woman Co-ingestions: Chlorthalidone 2250 mg Lisinopril unknown amount	Atenolol 4500 mg	Intervention: Oesophageal gastro duodenoscopy removal of pill fragments at 16 h Co-interventions: Fluid boluses, glucagon bolus and infusion, norepinephrine infusion, phenylephrine infusion, epinephrine infusion, Impella device, ECMO, CVVHDF	Mortality: No Morbidity: ARDS Day 6-7 Haemodynamic improvement: NR Secondary outcomes: NR Adverse effects: NR
Source	Subject and patient characteristics	Type of BB poisoning	Intervention and co-interventions	Results
Catecholamines, inotropes and vasopressors				
Harley et al. [67]	Subject: 57-year-old male Comorbidities: HT, IHD/PCI 2 years previously Co-ingestions: Diltiazem SR 960 mg, nifedipine modified release. 120 mg State at arrival: Unconscious, HR 48 BP 60/40	Bisoprolol 40 mg	Intervention: Epinephrine 0.1 mg boluses until BP > 90/60 and HR > 50 and epinephrine infusion up to 2 mcg/kg/min Co-interventions: Intubation/ventilation, SDAC, calcium chloride (no effect), CVVHD for acute renal failure	Mortality: No Morbidity: Acute renal failure, lactic acidosis, ST segment depression on ECG but no enzyme rise Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Amundson and Brodine [69]	Subject: 18-year-old male Comorbidities: Depression Co-ingestions: Nil State at arrival: Unconscious, seizure, gasping respirations, palpable pulse but no BP	Propranolol 2.4 g	Intervention: Dobutamine 8 mcg/kg/min and dopamine 20 mcg/kg/min in addition to epinephrine infusion up to 2 mcg/kg/min Co-interventions: Intubation/ventilation, atropine, isoprenaline, glucagon, lavage and SDAC, cardiac pacing, MAST suit	Mortality: Died Morbidity: Pulmonary oedema, low SVR Haemodynamic improvement: No Secondary outcomes: N/A Adverse effects: NR

(continued)

Continued.

Source	Subject and patient characteristics	Type of BB poisoning	Intervention and co-interventions	Results
Jones et al. [87]	Subject: 60-year-old male Comorbidities: Angina Co-ingestions: Nil State at arrival: Cardiac arrest	Propranolol 9.6 g	Intervention: Dopamine, Epinephrine isoprenaline (doses not stated)	Mortality: Died Morbidity: N/A Haemodynamic improvement: No Secondary outcomes: N/A Adverse effects: NR
Bourg et al. [88]	Subject: 26-year-old female Comorbidities: Grave's disease Co-ingestions: ethanol, propylthiouracil State at arrival: coma, pulse 40 (sinus), BP 60	Propranolol (dose not known – "whole bottle")	Intervention: Isoprenaline 120 mcg/min, dopamine 6.66 mg/min, norepinephrine Co-interventions: Intubation/ventilation, glucagon bolus and infusion, atropine, naloxone, lavage and SDAC, lignocaine for VT, bicarbonate	Mortality: No Morbidity: VT, seizure Haemodynamic improvement: Yes to glucagon but No to the catecholamines Secondary outcomes: NR Adverse effects: NR
Litman and Zerngast [152]	Subject: 11-year-old female Comorbidities: congenital myelomeningocele, State at arrival: Having spinal fusion for scoliosis. Pre-op: HR 100 BP 90/60. Given IV esmolol for controlled hypotension during surgery Within minutes: progressive decrease in HR and BP to asystole	Esmolol 12.5 mg IV over 1-2 min	Intervention: Ephedrine 50 mg, epinephrine 2.1 mg Co-interventions: CPR 15 min, atropine, calcium chloride 500 mg	Mortality: No Morbidity: Nil Haemodynamic improvement: Yes to calcium chloride and CPR but no to the catecholamines Secondary outcomes: NR Adverse effects: NR
Bouchard et al. [78]	Subject: 84-year-old male Comorbidities: Alzheimer's, HT Co-ingestions: simvastatin 300 mg State at arrival: Pale, BP 80/40 HR 45 RR 18 ECG – complete heart block with junctional escape	Carvedilol 375 mg	Intervention: Dopamine 10 mcg/kg/min Co-interventions: 1 l saline, lavage, SDAC, calcium chloride, glucagon boluses	Mortality: No Morbidity: Nil Haemodynamic improvement: Yes to both glucagon and dopamine Secondary outcomes: NR Adverse effects: NR
Bourguine et al. (abstract) [89]	Subject: 43-year-old female Comorbidities: Nil Co-ingestions: paracetamol, ethanol State at arrival: Confused, BP 90/60 HR 50 RR 18	Propranolol ('massive')	Intervention: Dobutamine and norepinephrine Co-interventions: IV NAC, ECMO	Mortality: No Morbidity: Hepatotoxicity Haemodynamic improvement: Yes to ECMO and dobutamine/norepinephrine Secondary outcomes: NR Adverse effects: NR
Halloran et al. [90]	Subject: 47-year-old female Comorbidities: HT, depression Co-ingestions: Nil State at arrival: Seizure, apnoeic, HR 50 BP 70/40	Propranolol	Intervention: Isoprenaline (32 mcg/min), dopamine (15 mcg/kg/min) and norepinephrine (12 mcg/min) Co-interventions: Intubation/ventilation, lavage, IV fluid, atropine	Mortality: No Morbidity: Aspiration pneumonitis Haemodynamic improvement: Yes to norepinephrine but No to isoprenaline and dopamine Secondary outcomes: NR Adverse effects: NR
Hicks and Rankin [153]	Subject: 43-year-old female Comorbidities: Nil Co-ingestions: Prednisolone 20–30 mg, doxepin 525 mg, azathioprine 4–6 tablets, temazepam 140 mg, 10–20 tablets GTN State at arrival: GCS 5, shallow breaths, HR 80 BP 60	Labetalol 5.6–7.0 g	Intervention: High dose epinephrine (5–20 mg boluses and 1333 mcg/min), dopamine (60 mg/h), isoprenaline 200 mcg boluses, dobutamine 250 mg Co-interventions: Intubation/ventilation, atropine, bicarbonate, SDAC, glucagon, calcium	Mortality: No Morbidity: Acute kidney injury, metabolic acidosis Haemodynamic improvement: Yes to high dose epinephrine Secondary outcomes: NR Adverse effects: NR
Hopkins et al. [91]	Subject: 21-year-old female Comorbidities: Nil Co-ingestions: Amlodipine 280 mg, olanzapine 560 mg State at arrival: GCS 13, HR 80 BP 90/45	Propranolol 6.4 g	Intervention: epinephrine (10–100 mcg boluses and 1–4 mg/kg/h) Co-interventions: Intubation/ventilation, IV fluid, HIET (10 U/kg bolus, 1 U/kg/h), ILE (1.5 ml/kg × 2, 0.25 ml/kg/min), delayed endoscopy and removal of pharmacobezoar	Mortality: No Morbidity: Nil Haemodynamic improvement: Yes to HIET, epinephrine and ILE combination Secondary outcomes: NR Adverse effects: NR

(continued)

Continued.

Source	Subject and patient characteristics	Type of BB poisoning	Intervention and co-interventions	Results
Isbister et al. [154]	Subject: 90-year-old female Comorbidities: COPD, AF, tachy-brady syndrome, PPM, on escalating therapeutic doses of metoprolol over 2 weeks Co-ingestions: Nil State at arrival: Lethargic, HR 100 AF, BP 83/53, RR 30, SpO ₂ < 80% on O ₂ , glucose 3.0 mmol/l	Metoprolol	Intervention: Norepinephrine 15–18 mcg/min, epinephrine 4 mcg/min, dobutamine 500 mcg/min Co-interventions: IV fluid, HIET (50 U bolus, 50–500 U/h), vasopressin, bicarbonate, intubation/ventilation	Mortality: No Morbidity: Acute renal failure, lactic acidosis, ischaemic hepatitis, disseminated intravascular coagulopathy Haemodynamic improvement: Yes to combination of all therapies Secondary outcomes: NR Adverse effects: NR
Kalman et al. [155]	Subject: 55-year-old male Comorbidities: Nil Co-ingestions: Verapamil 2.8 g State at arrival: pale, HR 20 SBP 40–50	Atenolol 1.2 g	Intervention: prenalterol 20–40 mg, dobutamine 52 mcg/kg, isoprenaline 0.4 mcg/kg/min, epinephrine 0.5 mg bolus and 2–2.5 mcg/kg/min, dopamine 33–50 mcg/kg/min, norepinephrine 0.3–0.5 mcg/kg/min Co-interventions: IV fluid, calcium gluconate, atropine	Mortality: No Morbidity: Lactic acidosis, mild hyperglycaemia Haemodynamic improvement: Yes to combination of all therapies Secondary outcomes: NR Adverse effects: NR
Lewis et al. [92]	Subject: 24-year-old female Comorbidities: Nil Co-ingestions: Trimipramine 625 mg State at arrival: Unconscious, HR 60 BP unrecordable, RR 30	Labetalol 7.2 g Acebutolol 9.6 g	Intervention: isoprenaline 30–1660 mcg/min, dopamine 50–200 mcg/kg/min Co-interventions: IV fluid, atropine, intubation/ventilation, lavage, calcium, glucagon	Mortality: No Morbidity: Metabolic acidosis, ischaemic bowel Haemodynamic improvement: Yes to combination of all therapies Secondary outcomes: NR Adverse effects: NR
Markota et al. [74]	Subject: 66-year-old male Comorbidities: Nil Co-ingestions: Amlodipine 300 mg, doxazosin 200 mg, nifedipine 600 mg, torasemide 150 mg, ASA 4 g, ibuprofen 5 g State at arrival: Unconscious, HR 30 BP 45/30	Bisoprolol 450 mg	Intervention: Epinephrine 0.1 mg boluses × 4, norepinephrine 0.5–14 mcg/kg/min, dobutamine 10 mcg/kg/min Co-interventions: IV fluid, calcium gluconate, glucagon, intubation/ventilation, lavage, SDAC, HIET (1 U/kg bolus, 1 U/kg/h), ILE (100 ml over 5 min, 150 ml over 60 min), vasopressin	Mortality: No Morbidity: Lactic acidosis Haemodynamic improvement: Yes to combination of all therapies Secondary outcomes: NR Adverse effects: NR
Salzberg and Gallagher [85]	Subject: 20-year-old female Comorbidities: Thyrotoxicosis Co-ingestions: Nil State at arrival: Awake, HR 110 BP 150/100 RR 24 but deteriorated soon after	Propranolol 2 g	Intervention: isoprenaline 4–20 mcg/min, dopamine 0.5–5 mg/min Co-interventions: IV fluid, atropine, ipecac, glucagon	Mortality: No Morbidity: Nil Haemodynamic improvement: Yes to glucagon but No to isoprenaline and dopamine Secondary outcomes: NR Adverse effects: NR
Tai et al. [84]	Subject: 23-year-old female Comorbidities: 20/40 gestation Co-ingestions: Nil State at arrival: Asystole	Metoprolol 15.2 g	Intervention: epinephrine boluses (5 mg total), epinephrine 600 mcg/h, dopamine 10 mcg/kg/min Co-interventions: CPR, calcium gluconate, intubation/ventilation, lavage, glucagon	Mortality: No Morbidity: Pulmonary oedema, aspiration pneumonia, foetal death in utero Haemodynamic improvement: Yes to glucagon in addition to the epinephrine and dopamine Secondary outcomes: NR Adverse effects: NR
Atropine Amundson and Brodine [69]	Subject: 18-year-old man	Propranolol 2.4 g	Intervention: Atropine bolus Co-interventions: Gastric lavage activated charcoal 100 g q4H, transvenous pacing, dobutamine, dopamine, military anti-shock trousers, isotonic crystalloid. Levarterenol, epinephrine, naloxone, calcium chloride	Mortality: Died after 9 h Morbidity: Early noncardiogenic pulmonary oedema, bronchorrhea, bronchospasm, vasodilatory shock Haemodynamic improvement: HR improved with atropine, but not BP Secondary outcomes: N/A Adverse effects: NR
Abbasi et al. [68]	Subject: 15-year-old girl	Atenolol 500 mg	Intervention: Atropine 0.5 mg bolus then had 11 subsequent boluses post admission Co-interventions: Gastric lavage, atropine boluses, glucose infusion, activated charcoal	Mortality: No Morbidity: NR Haemodynamic improvement: HR increased from 44 to 60, BP from 90/60 to 132/65 Had 11 further boluses of atropine 0.5 mg to maintain HR 50 and BP 90/60 Secondary outcomes: NR Adverse effects: NR

(continued)

Continued.

Source	Subject and patient characteristics	Type of BB poisoning	Intervention and co-interventions	Results
Lewis et al. [92]	Subject: 24-year-old female Co-ingestions: Trimipramine 625 mg	Acebutolol 9.6 g Labetalol 7.2 g	Intervention: Atropine 1 mg Co-interventions: Isoproterenol infusion, dopamine infusion 200 mcg/kg/min, calcium chloride 5% 10 ml × 3, glucagon 10 mg × 3	Mortality: No Morbidity: Re-admitted 26 days later for bowel obstruction secondary to ischaemic bowel necrosis Haemodynamic improvement: NR Secondary outcomes: N/A Adverse effects: NR
Salzberg and Gallagher [85]	Subject: 20-year-old female Comorbidities: Thyrotoxicosis	Propranolol 200 mg	Intervention: Atropine 1 mg Co-interventions: Syrup of ipecac 30 ml, Dextrose 25 g, Naloxone 0.8 mg, Isoproterenol infusion, Dopamine infusion, Glucagon 3 mg	Mortality: No Morbidity: Seizures × 3 Haemodynamic improvement: NR Secondary outcomes: NR Adverse effects: NR
Calcium Tai et al. [84]	Subject: 23-year-old female Comorbidities: 20 weeks pregnant Status on arrival: Cardiac arrest on hospital arrival	Metoprolol 15.2 g	Intervention: Calcium gluconate 3 g Co-interventions: Adrenaline 5 mg, Adrenaline infusion, Dopamine infusion, Glucagon 1 mg, Gastric Lavage	Mortality: No Morbidity: Pulmonary oedema, bilateral aspiration pneumonia, foetal Yes in utero Haemodynamic improvement: NR Secondary outcomes: NR Adverse effects: NR
Fung et al. [79]	Subject: 45-year-old female Comorbidities: Hypertension, hyperlipidaemia and renal impairment Co-ingestions: Nifedipine XR 1120 mg, Prazosin 84 mg, Glipizide 2240 mg and Fluvastatin 280 mg	Metoprolol 4200 mg	Intervention: Calcium chloride 1 g × 2 Co-interventions: Glucagon 3 mg then infusion, high dose insulin euglycaemic therapy	Mortality: No Morbidity: Nil Haemodynamic improvement: NR Secondary outcomes: NR Adverse effects: NR
Jones [93]	Subject: 42-year-old female Status on arrival: Accidental overdose, hypotensive and bradycardic	Metoprolol 800 mg/day for 3 days	Intervention: Calcium chloride unspecified dose Co-interventions: Vasopressors and chronotropic agents, transthoracic pacemaker	Mortality: No Morbidity: Nil Haemodynamic improvement: NR Secondary outcomes: NR Adverse effects: NR
Henry et al. [70]	Subject: 21-year-old female Co-ingestions: Verapamil unknown amount Status on arrival: Cardiogenic shock	Atenolol unknown amount	Intervention: IV 1 g calcium chloride Co-interventions: 3 l normal saline, Isoproterenol infusion, Gastric lavage, Activated charcoal, Magnesium sulphate	Mortality: No Morbidity: Nil Haemodynamic improvement: Blood pressure improved following calcium chloride administration Secondary outcomes: NR Adverse effects: NR
O'Grady et al. [94]	Subject: 50-year-old female Co-ingestions: Paracetamol, Clonazepam, Desipramine-unknown amounts	Atenolol unknown amount	Intervention: Calcium chloride 1 g × 2 Co-interventions: Normal saline boluses, Atropine, Ephedrine, Dopamine infusion, Adrenaline infusion, Glucagon 11 mg in total	Mortality: No Morbidity: Nil Haemodynamic improvement: Blood pressure improved following calcium chloride administration (twice) Secondary outcomes: NR Adverse effects: NR
Pertoldi et al. [71]	Subject: 20-year-old female Co-ingestions: Hydrochlorothiazide, Fluoxetine, Chlordesmetildiazepam-unknown amounts	Atenolol 1.8–2.5 g	Intervention: Calcium chloride 1 g × 3 with infusion Co-interventions: Gastric emptying, Activated charcoal, Cathartic, Flumazenil, Potassium chloride, Intravenous fluids, Epinephrine infusion, Magnesium sulphate, Glucagon, Temporary transvenous pacemaker, Enoximone infusion	Mortality: No Morbidity: Acute, non-oliguric renal failure Haemodynamic improvement: Blood pressure improved following calcium chloride administration Secondary outcomes: NR Adverse effects: NR
Insulin Barton et al. [72]	Subject: 59-year-old male (100 kg) Comorbidities: Methamphetamine use, paroxysmal AF, non-ischaemic cardiomyopathy, severe mitral regurgitation State at arrival: Unstable – BP 74/43, HR 90 with progression soon after to	Metoprolol tartrate 7.5 g (75 mg/kg)	Intervention: 1 U/kg (100 IU) bolus with 100 ml 50% dextrose then infusion of 100 IU/h with 10% dextrose at 200 ml/h post ROSC Required three boluses and up-titration of infusion to 250 IU/h for 22 h and was ceased after all other vasoactive agents had been discontinued	Mortality: No Morbidity: Nil listed Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR

(continued)

Continued.

Source	Subject and patient characteristics	Type of BB poisoning	Intervention and co-interventions	Results
	bradycardia (50 s) and worsening hypotension followed by cardiac arrest (PEA)		Co-interventions: <i>Prior to arrest</i> – atropine 1.5 mg, glucagon 6 mg, dopamine infusion 10 µg/kg/min. Intubation with 100 mg ketamine and 150 mg suxamethonium. <i>During arrest</i> – 1 mg adrenaline and noradrenaline infusion 0.1 µg/kg/min. 1.5 ml/kg (150 ml) 20% ILE. <i>Post arrest</i> – Calcium gluconate 3 g × 2, second bolus 150 ml ILE, adrenaline 0.15–0.2 µg/kg/min infusion (norad ceased), gastric lavage, 50 g activated charcoal, KCl and MgSO ₄ replacement 6 hourly	
Cole et al. [161]	Subject: 50-year-old female Comorbidities: NR Co-ingestions: Bupropion XR 150 mg State at arrival: Unstable. Haemodynamics listed post fluid resuscitation – HR 40 MAP 30	Metoprolol 4 g	Intervention: 10 U/kg/h insulin Co-interventions: IV fluids, 41 mEq Ca ²⁺ , 10 mg glucagon, 1.5 ml/kg 20% ILE, catecholamines (dose not listed), transvenous pacemaker, intra-aortic balloon pump	Mortality: Yes at 4 days post admission Morbidity: Multi-organ failure Haemodynamic improvement: No – patient had ongoing deterioration and cardiac arrest following insulin. Requiring 4 min CPR and ALS prior to ROSC Secondary outcomes: N/A Adverse effects: NR
Cole et al. [161]	Subject: 53 years old male Comorbidities: None provided Co-ingestions: Diltiazem SR 120 mg State at arrival: Unstable – HR 30 and MAP 40 despite fluid resuscitation	Propranolol 20 mg	Intervention: 10 U/kg/h insulin infusion Co-interventions: IV fluids, 32 mEq of Ca ²⁺ , 2 mg of adrenaline, 2 mg atropine, dopamine infusion (dose not listed), 1.5 ml/kg 20% ILE, transvenous pacemaker, IABP	Mortality: Yes at 7 days post admission Morbidity: Multi system organ failure Haemodynamic improvement: No – patient had ongoing deterioration and cardiac arrest following insulin. Required 5 min CPR and ALS for ROSC. Secondary outcomes: N/A Adverse effects: NR
Czerwinka and Heim et al. [73]	Subject: 33-year-old female Comorbidities: None Co-ingestions: Zolpidem, ethanol State at arrival: Unstable – had been involved in MVA, level 1 trauma, I + V GCS 3 with HR 54 BP 60/30	Atenolol – dose not listed	Intervention: Dose utilised of insulin not listed Co-interventions: 10 litres fluid resuscitation, 8 units PRBC, 1 g TXA & infusion, 1 unit FFP, dopamine, noradrenaline, activated charcoal, acetylcysteine, calcium, glucagon, HIET and ILE (doses not listed)	Mortality: No Morbidity: Possible ischaemic bowel (? discharged home same day) Haemodynamic improvement: Not specifically listed, but assumed yes Secondary outcomes: NR Adverse effects: NR
Doecker et al. [40] (Two patients but only one with BB toxicity)	Subject: 35-year-old male Comorbidities: HTN, T1DM Co-ingestions: Amlodipine, verapamil State at arrival: Unstable – bradycardic arrest 20 bpm, no measurable blood pressure	Metoprolol unknown amount and time (but estimated to be 1.18 g)	Intervention: Insulin 1 unit/kg (83 unit) bolus then 80 unit/h infusion with 10% dextrose at 100 ml/h. Unsuccessful wean 7 h after presentation but resumed due to hypotension. Infusion was weaned after 29 h Co-interventions: Atropine 2 g, dopamine 20 µg/kg/min and adrenaline 0.05 µg/kg/min and glucagon infusions 1 mg/h. 1.5 ml/kg (130 ml) ILE bolus and infusion of 0.25 ml/kg/min for 60 min	Mortality: No Morbidity: Nil Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: One episode of hypoglycaemia noted whilst on HIET and 3 further hypoglycaemic episodes 24 h after HIET
Escajeda et al. [112]	Subject: 47-year-old female Comorbidities: NR Co-ingestions: Nil State at arrival: Unstable – bradycardic, hypotensive, comatose	Metoprolol 10 g (100 × 100 mg)	Intervention: 'High dose regular insulin' 250 units IV (no further dosing or time frame listed) Co-interventions: I + V (ketamine 80 mg/roc 80 mg), glucagon 14 mg, adrenaline 0.1 mcg/kg/min, dobutamine 10 mcg/kg/min, adrenaline 5 mg (ALS) and vasopressin 40 units	Mortality: No Morbidity: Nil Haemodynamic improvement: Yes – post initiation of VA-ECMO Secondary outcomes: NR Adverse effects: NR

(continued)

Continued.

Source	Subject and patient characteristics	Type of BB poisoning	Intervention and co-interventions	Results
Fung et al. [79]	Subject: 45-year-old female Comorbidities: Hypertension, diabetes, hyperlipidaemia, renal impairment Co-ingestions: Nifedipine SR 1.12 g, prazosin 84 mg, gliclazide 2.24 g, fluvastatin 280 mg State at arrival: Unstable – GCS 14, BP 79/54, HR 55, ECG 1 st degree HB	Metoprolol 4.2 g	IV, 1.5 ml/kg 20% ILE then 0.25 ml/kg for 30 min and VA-ECMO Intervention: Actrapid 6 IU/h (0.1 IU/kg/h) plus dextrose 6 g/h, continued for 23 h Co-interventions: IV fluid (1 l), 9 mg glucagon, 1 g calcium chloride × 2, glucagon infusion 9 mg/h, 4 doses 50 g AC	Mortality: No Morbidity: Acute on chronic kidney injury, resolved by discharge Haemodynamic improvement: Yes – although appeared to correlate with glucagon rather than insulin as per the case report Secondary outcomes: NR Adverse effects: NR
Hasin et al. [158]	Subject: 59-year-old female Comorbidities: Hypertension Co-ingestions: Verapamil 3 g, enalapril (6 tablets) State at arrival: Unstable – SBP 70, HR 60, lethargic. Syncope prior to presentation.	Therapeutic dose of metoprolol	Intervention: 1–2 units/h (started 48 h post admission) with 5% glucose 140 ml/h continued for 38 h. Co-interventions: 3T fluid, dopamine 10–40 mg/h, calcium gluconate 10 ml 10% × 5, glucagon 4 mg bolus then 1–3 mg/h, noradrenaline	Mortality: No Morbidity: Acute kidney injury Haemodynamic improvement: Haemodynamic improvement 3 hours post institution of HIET Secondary outcomes: NR Adverse effects: NR
Hopkins et al. [91]	Subject: 21-year-old person (gender not disclosed) Comorbidities: Nil Co-ingestions: Amlodipine 280 mg (28 × 10 mg), Olanzapine 560 mg (56 × 10 mg) State at arrival: Unstable – BP 90/45 HR 80, GCS 13	Propranolol 6.4 g (160 × 40 mg)	Intervention: 10 units/kg/h IV insulin followed by infusion of 1 unit/kg/h Co-interventions: IV resuscitation, glucagon (10 mg/kg), 1 + V, adrenaline boluses then infusion 1 mg/kg/h, ILE 1.5 l/kg bolus × 2 then maintenance 0.25 ml/kg/min for 2 days (total 425 ml 20%). 48 h post admission due to lack of improvement, endoscopy undertaken with removal 200 ml tablets	Mortality: No Morbidity: Nil Haemodynamic improvement: Temporary stabilisation post insulin bolus but remained hypotensive. Main improvement noted improvement post endoscopy Secondary outcomes: NR Adverse effects: NR
Isbister et al. [154]	Subject: 90-year-old female Comorbidities: COPD, AF with tachy/brady syndrome Co-ingestions: Nil State at arrival: Unstable – HR 120, unrecordable BP, hypoglycaemic 3.0	Therapeutic dosing as guided by cardiologist – Metoprolol initially 25 mg BD, then 50 mg TDS (7 days prior to admission), then 100 mg mane, 50 mg midi, 100 mg nocte (day prior to admission)	Intervention: Insulin 50 units (1 unit/kg) bolus followed by infusion 50 U/h with 50% dextrose infusion 20 ml/h initially. After subsequent haemodynamic deterioration, was increased to 3 U/kg/h (150 U/h) then 10 U/kg/h (500 U/h) with 35 ml/h 50% dextrose. This was weaned over days 3–4 of admission. Co-interventions: Fluid resuscitation, noradrenaline 15 µg/min, NaHCO ₃ 8.4% 100 ml, dobutamine, vasopressin, adrenaline and haemofiltration	Mortality: No Morbidity: Renal failure requiring haemofiltration Haemodynamic improvement: Yes – post escalation of insulin rate, but multiple other therapies being concurrently used Secondary outcomes: NR Adverse effects: NR
Markota et al. [74]	Subject: 66-year-old male Comorbidities: Depression and HTN Co-ingestions: Amlodipine 300 mg, Doxazosin 200 mg, Nifedipine 600 mg, Torasemide 150 mg, Acetylsalicylic acid 4000 mg, Ibuprofen 5000 mg State at arrival: Unstable – unconscious, hypotensive BP 45/30 and HR 30	Bisoprolol 450 mg	Intervention: 1 IU/kg bolus followed by infusion of 1 IU/kg/h continued until day 3 Co-interventions: Adrenaline, 10 ml 10% calcium gluconate × 4, 15 mg glucagon, 1500 ml N/Saline, noradrenaline & dobutamine infusions, gastric lavage, 50 g activated charcoal, 100 ml 20% ILE bolus then 150 ml over 1 h, vasopressin infusion was also added	Mortality: No Morbidity: Metabolic acidosis, acute kidney injury Haemodynamic improvement: Reported improvement post HIET, but note that vasopressin also needed to be added due to increasing noradrenaline requirements Secondary outcomes: NR Adverse effects: NR
Montague et al. [159]	Subject: 7-month-old female Comorbidities: Haemangioma, IUGR Co-ingestions: Nil State at arrival: Unstable – cyanotic, bradycardic, hypotensive	Propranolol 36 mg for three doses (TDS) i.e., 17 mg/kg in a day	Intervention: 1 unit/kg (0.82 U/kg) bolus with 1 g/kg dextrose then infusion 1 U/kg/h and dextrose 0.5 g/kg/h which was continued for 3 h Co-interventions: Fluid resus 20 ml/kg, 20% ILE 0.25 ml/kg bolus then 0.0125 ml/kg 20% ILE over two hours with a second 0.2 ml/kg bolus when haemodynamics not improved	Mortality: No Morbidity: Nil Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: One episode of hypoglycaemia

(continued)

Continued.

Source	Subject and patient characteristics	Type of BB poisoning	Intervention and co-interventions	Results
Murphy et al. [82]	Subject: 72-year-old female Comorbidities: Hypertension Co-ingestions: Hydrochlorothiazide 1.5 g State at arrival: Unstable – GCS 14 HR 40 SBP 90	Metoprolol 3.75 g	Intervention: Insulin 85 unit bolus with 50 ml of 50% dextrose, then infusion of 40 units/h with dextrose 10% at 100 ml/h weaned by 24 h post ingestion Co-interventions: 150 g AC on arrival to ED, glucagon 5 mg IV bolus followed by 5 mg/h, calcium chloride (no dose listed)	Mortality: No Morbidity: Nil Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: Hypokalaemia – resolved by 6 h into admission
Paetow et al. [83]	Subject: 51-year-old male Comorbidities: Not listed Co-ingestions: Amlodipine unknown quantity State at arrival: Unstable – obtunded, SBP 50 s HR 20 s	Metoprolol 1000 mg (40 × 25 mg)	Intervention: 1 U/kg/h infusion (no bolus) increased to 10 U/kg/h over 3 h due to ongoing instability Co-interventions: Adrenaline boluses, 9 g calcium gluconate, I + V, adrenaline infusion, 50 g AC	Mortality: No Morbidity: NR Haemodynamic improvement: Yes, but in collaboration with commencing adrenaline infusion Secondary outcomes: NR Adverse effects: NR
Page et al. [96]	Subject: 45-year-old female (55 kg) Comorbidities: SVT Co-ingestants: Ethanol State at arrival: Unstable GCS 14, HR 54 BP 86/61	Metoprolol 5 g (100 × 50 mg)	Intervention: Four insulin boluses of 1–2 U/kg were administered to a total of 300 IU over 55 min with boluses 50% dextrose then infusion at 200 U/h, which was titrated up to 500 U/h (10 IU/kg). Two additional boluses required. Weaned at 14 h and ceased 17 h post ingestion Co-interventions: Atropine 3 mg, 3 l fluid resuscitation, isoprenaline boluses then infusion, 1 mg metaraminol	Mortality: No Morbidity: Nil Haemodynamic improvement: Noted to stabilise with insulin boluses and high dose (10 U/kg) infusion Secondary outcomes: NR Adverse effects: No hypoglycaemia reported, potassium maintained low normal 3.5 mmol/l
Qureshi et al. [75]	Subject: 48-year-old male Comorbidities: T2DM, HTN Co-ingestions: Amlodipine, hydralazine, hydrochlorothiazide, lisinopril, doxazosin- unknown dose State at arrival: Unstable – unresponsive, BP 53/21, pulse 89	Carvedilol (unknown dose)	Intervention: 135 U bolus (1 IU/kg) and then infusion at 300 U/h (2 U/kg/h) with dextrose and titrated up to 900 U/h (6 U/kg/h) over 3 h Co-interventions: I + V, IV fluids, noradrenaline, dopamine, vasopressin, adrenaline, activated charcoal, calcium chloride, glucagon, sodium bicarbonate and ILE (doses not listed). IABP	Mortality: Yes Morbidity: Iatrogenic left superficial femoral artery dissection and subsequent retroperitoneal haemorrhage > haemorrhagic shock (eventual cause of Yes) Haemodynamic improvement: Yes – significant improvement following commencement of HIET BP 55/24 > BP 91/54 and was extubated at day 8, however, later died (as above) Secondary outcomes: N/A Adverse effects: No episodes of hypoglycaemia whilst on HIET
Seegobin et al. [76]	Subject: 54-year-old female Comorbidities: Hypertension Co-ingestions: Hydrochlorothiazide, doxazocin, amlodipine (unknown amounts) State at arrival: Unstable – BP 72/53, HR 50	Atenolol (unknown amount)	Intervention: 1 U/kg/h (no bolus) titrated to 4 U/kg/h with dextrose 10%. Post institution of 5 vasopressors, insulin titrated to 10 IU/kg/h Co-interventions: I + V, intravenous fluids, activated charcoal, glucagon 5 mg followed by infusion 5 mg/h, calcium gluconate 10% and atropine, noradrenaline, vasopressin, phenylephrine, adrenaline and dopamine titrated to max doses (not listed), dialysis, ILE	Mortality: No Morbidity: Pulmonary oedema Haemodynamic improvement: Significant improvement 2 h after high dose (10 IU/kg/h) commenced allowing for vasopressors to be weaned down/off. Secondary outcomes: NR Adverse effects: Hypoglycaemia noted after insulin was weaned, persisting for 48 h after insulin stopped requiring ongoing dextrose infusion for 48 h
Singh et al. [160]	Subject: 28-year-old person (gender not disclosed) Comorbidities: NR Co-ingestions: NR State at arrival: Unstable – bradycardic, hypothermic and hypotensive	Propranolol (dose not listed)	Intervention: 'Insulin drip' (information lacking regarding dose) Co-interventions: Naloxone, propofol and keppra (for seizure control), adrenaline, noradrenaline, dopamine and vasopressin, glucagon infusion, calcium gluconate	Mortality: No Morbidity: Nil reported Haemodynamic improvement: Yes, post commencing insulin Secondary outcomes: NR Adverse effects: NR

(continued)

Continued.

Source	Subject and patient characteristics	Type of BB poisoning	Intervention and co-interventions	Results
Stellpflug et al. [119]	Subject: 48-year-old male Comorbidities: Hypertension, paraplegia (wheelchair bound) Co-ingestants: Baclofen, diazepam State at arrival: Unstable – unresponsive GCS 3, BP 92/52, HR 72	Nebivolol	Intervention: 100 IU bolus then infusion commenced (initial dose not listed) titrated up to 21 U/kg/h (administrative dose error) which was then titrated back down (dose not listed) and continued for 36 h Co-interventions: 3 l IV fluid resuscitation, calcium, atropine 2 mg and 1 mg adrenaline & 100 ml bolus of ILE (during CPR then ROSC obtained), infusion of 0.25 ml/kg/min ILE for 1 h (1 l total)	Mortality: No Morbidity: Nil Haemodynamic improvement: No clear improvement listed (but stabilised) Secondary outcomes: NR Adverse effects: Nil hypoglycaemia
Stellpflug et al. [120]	Subject: 30-year-old female Comorbidities: HOCM with AICD Co-ingestions: Diltiazem, amiodarone State at arrival: Stable	Metoprolol (unknown dose)	Intervention: Insulin bolus 0.5 U/kg followed by infusion which was escalated over an hour to 10 U/kg/h Co-interventions: 4 l Normal Saline, 27 mEq calcium gluconate, ILE 20% 100 ml with infusion of 1.5 l over 1 h	Mortality: No Morbidity: Acute kidney injury Haemodynamic improvement: Yes – as per case report, noted after ILE given (85 min after HIET initiated) Secondary outcomes: NR Adverse effects: NR
Glucagon Garg et al. [77]	Subject: 32-year-old male Comorbidities: Type 1 Diabetes Mellitus Co-ingestions: Digoxin 25 mg State at arrival: Conscious, vitals stable on presentation bar BSL 58 mg/dl (3.2 mmol/l)	Propranolol 2 g	Intervention: 5 mg glucagon slow IV bolus then infusion at 5 mg/h for 19 h before reduction to 2 mg/h for another 21 h Co-interventions: Gastric lavage, IV dextrose, atropine 0.6 mg × 2 IV, charcoal haemoperfusion for 15 h	Mortality: No Morbidity: None Haemodynamic improvement: HR maintained above 50 bpm per no comment regarding blood pressure Secondary outcomes: NR Adverse effects: NR
Bouchard et al. [78]	Subject: 84-year-old male Comorbidities: Alzheimer's dementia, hypertension Co-ingestions: Simvastatin 300 mg State at arrival: Pale but alert, BP 70/52 mmHg, 70 bpm	Carvedilol 6.25 mg	Intervention: IV glucagon boluses 2–3 mg each Co-interventions: 2 l 0.9% normal saline bolus then 100 ml/h, 13 mEq calcium chloride, gastric lavage, activated charcoal, dopamine infusion 10 µg/kg/min	Mortality: No Morbidity: NR Haemodynamic improvement: BP improved to 100/60 mmHg lasting 30–45 min Secondary outcomes: NR Adverse effects: NR
Fung et al. [79]	Subject: 45-year-old female Comorbidities: Hypertension, diabetes mellitus, hyperlipidaemia, and renal impairment Co-ingestions: Nifedipine XR 1.12 g, prazosin 84 mg, glyclazide 2.24 g, fluvastatin 280 mg State at arrival: GCS 14/15, BP 79/54 mmHg, PR 55 bpm, BSL 8.6 mmol/l	Metoprolol 4.2 g	Intervention: IV glucagon 3 mg then glucagon infusion at 9 m/h commenced later on for 20 h Co-interventions: Normal saline, 10% calcium chloride (10 ml) × 2, insulin/dextrose, activated charcoal 50 g × 4	Mortality: No Morbidity: NR Haemodynamic improvement: Improvement of BP post bolus, improved BP after insulin/dextrose started (glucagon had been previously stopped) Secondary outcomes: NR Adverse effects: Vomiting post glucagon bolus
Love and Elshami [97]	Subject: 50-year-old male Comorbidities: NR Co-ingestions: Ethanol, lorazepam and chlorthalidone State at arrival: Asymptomatic	Atenolol 5 g	Intervention: Glucagon 5 mg bolus then infusion at 1 mg/h Co-interventions: None	Mortality: No Morbidity: None Haemodynamic improvement: BP improved with glucagon boluses but no values provided Secondary outcomes: NR Adverse effects: NR
DeLima et al. [98]	Subject: 50-year-old male Comorbidities: Depression, coronary artery disease, hypertension, asthma Co-ingestions: Nil State at arrival: Alert and cooperative with normal vitals	Atenolol 1 g	Intervention: Glucagon boluses (1 mg) plus infusion (dose not reported) Co-interventions: Gastric lavage, external pacemaker, 2 mg atropine, 1.9 l crystalloid, intubation, 60 µg/min isoproterenol, 13 µg/min norepinephrine, albuterol	Mortality: No Morbidity: None Haemodynamic improvement: Short-lived post bolus, unclear following institution of multiple other treatments Secondary outcomes: NR Adverse effects: NR
Fernandes and Daya [163]	Subject: 80-year-old female Comorbidities: Hypothyroidism, paroxysmal AF, angina Co-ingestions: None	Sotalol (prescribed dose 80 mg BD)	Intervention: 0.2 mg/min initially (max 3.5 mg), then additional 15 mg over 5 h for sinus arrest Co-interventions: None	Mortality: No Morbidity: Nil Haemodynamic improvement: NR Secondary outcomes: NR

(continued)

Continued.

Source	Subject and patient characteristics	Type of BB poisoning	Intervention and co-interventions	Results
Mansell [99]	State at arrival: Pale, HR 42 bpm, BP 122/60 Subject: 55-year-old female Comorbidities: Hypertension Co-ingestions: Amitriptyline State at arrival: Comatose, normal vitals otherwise	Propranolol	Intervention: 4 mg bolus glucagon then infusion Co-interventions: 0.9% saline	Adverse effects: One episode of nausea and vomiting Mortality: No Morbidity: Nil Haemodynamic improvement: Bolus caused increase in blood pressure however only lasted 10 min Secondary outcomes: NR Adverse effects: One episode hypoglycaemia (1.8 mmol/l) post cessation of infusion
O'Mahony et al. [100]	Subject: 16-year-old female Comorbidities: None Co-ingestions: None State at arrival: GCS 3, unrecordable BP, cool extremities, HR 68 bpm with weak pulses	Oxprenolol 8 g	Intervention: Glucagon 10 mg IV total (boluses) initially then 30 mg IV total (boluses later on) then infusion 10 mg/h with isoprenaline and dopamine Co-interventions: Gastric lavage post intubation, IV crystalloid, isoprenaline infusion 10 µg/min, dopamine 10 µg/kg/min	Mortality: No Morbidity: Nil Haemodynamic improvement: After 2 h of commencing glucagon infusion as well as other vasopressors Secondary outcomes: NR Adverse effects: NR
Agura et al. [136]	Subject: 40-year-old female Comorbidities: Hypertension Co-ingestions: Ethanol, antibiotics State at arrival: Unresponsive, hypotensive, BP 70 mmHg on palpation, HR 60 bpm	Propranolol (dose not reported)	Intervention: 10 mg IV glucagon then infusion post ROSC in ICU Co-interventions: IV fluids, atropine (1 mg), thiamine (100 mg), glucose (50 mg), naloxone (0.8 mg), isoproterenol 16–20 mcg/min, IV epinephrine 1 mg, sodium bicarbonate 54 MeQ, dopamine 15–20 mcg/kg/min	Mortality: No Morbidity: Respiratory arrest Haemodynamic improvement: Vague however appears hypotension dependent on norepinephrine not glucagon Secondary outcomes: NR Adverse effects: NR
Khan and Miller [103]	Subject: 28-year-old male Comorbidities: Nil Co-ingestions: Nil State at arrival: Cold, clammy, hypotensive (sBP 40 mmHg), HR 40 bpm	Propranolol 3 g	Intervention: 10 mg IV glucagon repeated twice on day 2 Co-interventions: Gastric lavage, atropine, isoprenaline 0.55–1.1 mcg/kg/min	Mortality: No Morbidity: Nil Haemodynamic improvement: On day 2 whilst on isoprenaline infusion, 30 min after second dose Secondary outcomes: NR Adverse effects: NR
Ward and Jones [102]	Subject: 62-year-old male Comorbidities: NR Co-ingestions: Diazepam State at arrival: CPR on arrival via ambulance, asystolic arrest	Oxprenolol	Intervention: Glucagon 10 mg IV + infusion 2 mg/h for 5 h Co-interventions: Isoprenaline infusion, IV adrenaline 0.3 mg, calcium gluconate 20 ml, sodium bicarbonate 100 ml, adrenaline infusion	Mortality: No Morbidity: Asystolic arrest Haemodynamic improvement: BP improved from 30 mmHg to 150 mmHg post 10 mg bolus within 60 s as per authors Secondary outcomes: NR Adverse effects: NR
Methylthioninium chloride (methylene blue) Aggarwal et al. [104]	Subject: 69-year-old female Comorbidities: Hypothyroidism, depression Co-ingestions: amlodipine State at arrival: Somnolent, HR 49 BP 69/35, vomiting	Atenolol	Intervention: methylene blue 1 mg/kg over 10 min then 1 mg/kg/h for 10 h Co-interventions: Intubation/ventilation, IV fluids, glucagon, CaCl, norepinephrine 1 mcg/kg/min, dopamine 20 mcg/kg/min, vasopressin 0.8 U/min, HIET, pacing, defibrillation	Mortality: No Morbidity: VF Haemodynamic improvement: Mildly improved with vasopressors; Yes with methylene blue Secondary outcomes: NR Adverse effects: NR
Kim et al. (abstract) [105]	Subject: 69-year-old female Comorbidities: Hypertension, depression Co-ingestions: Amlodipine, thioridazine State at arrival: HR 44 BP 67/40, ECG – sinus bradycardia	Atenolol	Intervention: Methylene blue 1 mg/kg over 10 min then 2 mg/kg/h for 5 h × 2 Co-interventions: IV fluids, glucagon, CaCl, norepinephrine, dopamine, vasopressin, HIET, pacing	Mortality: No Morbidity: Nil Haemodynamic improvement: Yes, with methylene blue Secondary outcomes: NR Adverse effects: NR
Laes et al. (abstract) [106]	Subject: 54-year-old male Comorbidities: Hypertension Co-ingestions: Amlodipine, valsartan, CO	Atenolol	Intervention: Methylene blue 2 mg/kg over 30 min then 0.75 mg/kg/h for 12 h Co-interventions: Intubation/ventilation, IV fluids 2 l,	Mortality: No Morbidity: ARF, pulmonary oedema, lactic acidosis Haemodynamic improvement: Yes, with methylene blue

(continued)

Continued.

Source	Subject and patient characteristics	Type of BB poisoning	Intervention and co-interventions	Results
	State at arrival: Normal vital signs		glucagon, Ca gluconate, vasopressin, phenylephrine, dopamine, epinephrine, HIET, norepinephrine, CVVHD, ILE bolus	Secondary outcomes: NR Adverse effects: NR
Laes et al. [107]	Subject: 61-year-old male Comorbidities: T2DM, Hypertension, depression Co-ingestions: Amlodipine 200 mg, valsartan 375 mg, CO State at arrival: Normal vital signs	Atenolol 500 mg	Intervention: Methylene blue 2 mg/kg over 30 min then 0.75 mg/kg/h Co-interventions: Intubation/ventilation, IV fluids 2 l, glucagon, Ca gluconate, vasopressin, phenylephrine, dopamine, epinephrine, HIET, norepinephrine, CVVHD, ILE bolus	Mortality: No Morbidity: ARF, pulmonary oedema, lactic acidosis Haemodynamic improvement: Yes, with methylene blue Secondary outcomes: NR Adverse effects: NR
Intravenous lipid emulsion (ILE) Barton et al. [72]	Subject: 59-year-old male Comorbidities: Methamphetamine abuse, AF, cardiomyopathy, severe MR Status at arrival: Unstable, hypotension, cyanotic, diaphoretic, AF	Metoprolol 7.5 g	Intervention: ILE 1.5 ml/kg 20% (2 boluses) Co-interventions: Atropine, glucagon, dopamine infusion, noradrenaline, HIET, gastric lavage, AC	Mortality: No Morbidity: PEA arrest Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Biary et al. (abstract) [108]	Subject: 78-year-old female Comorbidities: AF Status at arrival: Asystole, HR 37 with 6–10 s pauses	Propafenone 150 mg (not deliberate)	Intervention: 90 ml 20% ILE Co-interventions: Atropine, calcium, glucagon, ablation	Mortality: No Morbidity: Asystole Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Carr et al. (abstract) [109]	Subject: 31-year-old female Status at arrival: Hypotensive, HR 73	Carvedilol 875 mg	Intervention: 100 ml 20% ILE + 150 ml over 15 min Co-interventions: IV fluids, glucagon, Calcium Chloride, dopamine, adrenaline, HIET	Mortality: No Morbidity: Nil Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Cumpston et al. [110]	Subject: 55-year-old male Co-ingestions: Amlodipine – unknown amount Status at arrival: Altered conscious state with slurred speech, hypotensive, HR 57	Labetolol – unknown amount	Intervention: 0.25 ml/kg/min 20% ILE for 3 h, 0.17 ml/kg/min for 90 min, 0.004 ml/kg/min for 30 h Co-interventions: IV fluids, glucagon, Ca Glu, CaCl, noradrenaline, HDI	Mortality: No Morbidity: Nil Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Czerwonka and Heim (abstract) [73]	Subject: 33-year-old female Comorbidities: Nil Co-ingestions: Zolpidem, paracetamol – unknown if co-ingested Status at arrival: Post MVA to trauma centre, hypotensive, bradycardic 54, GCS 3 (negative trauma imaging)	Atenolol- unknown amount	Intervention: ILE, no dosing Co-interventions: IV fluids, PRBC, FFP, TXA, dopamine, noradrenaline, AC, NAC, calcium, glucagon, HIET	Mortality: No Morbidity: Nil Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Dean et al. [111]	Subject: 27-year-old female Status at arrival: Collapse, hypotensive, GCS 3	Propranolol 7 g	Intervention: 100 ml 20% ILE followed by 400 ml over 20 min Co-interventions: IV fluids, atropine, glucagon, HDI, lorazepam, phenytoin, isoprenaline, cardiac pacing, adrenaline, AC	Mortality: No Morbidity: Seizure Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Escajeda et al. [112]	Subject: 47-year-old male Comorbidities: Ethanol abuse, hypertension, depression, Status at arrival: Alert when ambulance arrived but became hypotensive With altered conscious state, bradycardia, hypotensive en-route to hospital	Metoprolol 10 g	Intervention: 1.5 ml/kg bolus and 0.25 ml/kg for 30 min 20% ILE Co-interventions: IV fluids, glucagon, adrenaline, dobutamine, adrenaline boluses, vasopressin, HDI, VA ECMO, cooled	Mortality: No Morbidity: Pneumonia, acute renal injury, transaminitis Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Hopkins et al. [91]	Subject: 21-year-old person – gender not disclosed Co-ingestions: Amlodipine 280 mg, olanzapine 560 mg Status at arrival: Hypotensive, HR 80, GCS 13	Propranolol 6.4 g	Intervention: 1.5 ml/kg 20% ILE and then 0.25 ml/kg/min for 2 days (total 425 ml) Co-interventions: IV fluids, glucagon, adrenaline, HDI, endoscopic removal of bezoars (200 ml)	Mortality: No Morbidity: Nil Haemodynamic improvement: Yes, but indirect, appeared to improve post endoscopy Secondary outcomes: NR Adverse effects: NR

(continued)

Continued.

Source	Subject and patient characteristics	Type of BB poisoning	Intervention and co-interventions	Results
Johnson-Arbor et al. [113]	Subject: 43-year-old female Comorbidities: Depression Co-ingestions: Tramadol 1.5 g, Zolpidem 200 mg, Alprazolam 15 mg Status on arrival: Found with agonal breathing, pulseless, hypotension, HR 81	Propranolol ?12 g	Intervention: 1.5 ml/kg 20% ILE × 2, followed by 0.25 ml/kg/min + another 1.5 ml/kg bolus Co-interventions: IV fluids, HDI, adrenaline, vasopressin, dobutamine, ECMO	Mortality: Yes Morbidity: Asystolic arrest Haemodynamic improvement: No Secondary outcomes: NR Adverse effects: Interference with lab results
Jovic-Stosic et al. [114]	Subject: 31-year-old female Comorbidities: Alcoholism Co-ingestions: Ethanol Status at arrival: Coma, HR 60, hypotensive, hypoxia	Propranolol 3.6 g	Intervention: 100 ml 20% ILE and 400 ml over 20 min + 200 ml at 10 ml/min Co-interventions: 5% dextrose, diazepam, atropine, glucagon, HIET, NaHCO ₃ , dobutamine	Mortality: No Morbidity: Seizure, broad complex tachycardia Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Macala et al. (abstract) [115]	Subject: 48-year-old female Co-ingestions: Clonidine 2.8 mg, Escitalopram 400 mg, Lorazepam 28 mg, Atorvastatin 560 mg, Quetiapine 5.6 g Status at arrival: HR 71, hypotensive, GCS 6	Propranolol 1.12 g	Intervention: 100 ml 20% ILE followed by 40 ml over 20 min Co-interventions: IV fluids, naloxone, glucagon, adrenaline, dopamine, vasopressin, HIET	Mortality: No Morbidity: Nil Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Markota et al. [74]	Subject: 66-year-old male Co-ingestions: Amlodipine 300 mg, Doxazosin 200 mg, Nifedipine 600 mg, Torasemide 150 mg, Acetylsalicylic acid 4 g, Ibuprofen 5 g Status at arrival: Unconscious, hypotensive, bradycardic	Bisoprolol 450 mg	Intervention: 100 ml 20% ILE and 150 ml over 60 min Co-interventions: Adrenaline, calcium gluconate, glucagon, fluids, noradrenaline, dobutamine, HIET, vasopressin, gastric lavage, AC	Mortality: No Morbidity: Severe metabolic acidosis, renal impairment Haemodynamic improvement: Yes Secondary outcomes: Extubated day 18, discharged from ICU day 25, discharged home neurologically intact day 62 Adverse effects: NR
Orthober et al. [116]	Subject: 36-year-old man Comorbidities: Bipolar affective disorder, depression, hypertension Co-ingestions: Methamphetamine, 'moonshine', 4–6 tabs promethazine, 5–8 tabs pregabalin, 12–15 tabs meclizine Status at arrival: Altered conscious (lethargy, slurring) → unresponsive, hypotensive, bradycardia -transferred → cardiac arrest in tertiary ED	Metoprolol 10–12 tablets	Intervention: ILE ?dose Co-interventions: IV fluids, glucagon, isoproterenol, adrenaline, atropine, HIET, NaHCO ₃ , calcium chloride, phenylephrine, NAC	Mortality: No Morbidity: Blind (light perception only) due to severe hypoxia, renal failure, adrenal insufficiency, hepatic ischaemia. Required tracheostomy, haemodialysis and gastrotomy. Haemodynamic improvement: Yes Secondary outcomes: Left hospital against medical advice Adverse effects: NR
Pasha et al. (abstract) [117]	Subject: 32-year-old female Comorbidities: Depression Co-ingestions: Amlodipine 30 tablets Status at arrival: Unresponsive, hypotensive	Carvedilol 130 tablets	Intervention: ILE ?dose Co-interventions: IV fluids, calcium gluconate, glucagon, HIET, 10% dextrose, adrenaline, noradrenaline, dopamine, vasopressin	Mortality: No Morbidity: ARDS, pancreatitis Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: Lab interference
Rodriguez et al. [118]	Subject: 26-year-old male Comorbidities: Hypertension, renal transplant Co-ingestions: Amlodipine, Lisinopril Status at arrival: Hypotensive, bradycardia	Metoprolol	Intervention: 1.5 ml/kg 20% ILE × 2, 0.25 ml/kg/min (20 ml/min). Total 5 h infusion (6200 ml) Co-interventions: IV fluids, glucagon, calcium gluconate, 50% dextrose, 'regular insulin', dopamine, adrenaline, noradrenaline, vasopressin, CVVHDF Required centrifugal plasma exchange to remove ILE	Mortality: Yes Morbidity: Hypoxia, renal failure, hyperkalaemia, acidaemia, pulmonary oedema Haemodynamic improvement: No Secondary outcomes: NR Adverse effects: Interference with CVVHDF
Stellpflug et al. [119]	Subject: 48-year-old male Comorbidities: Hypertension, paraplegic Co-ingestions: Baclofen 2.7 g, Diazepam 75 mg Status at arrival: Unresponsive, GCS 3, Hypotensive, bradycardia	Nebivolol 300 mg	Intervention: 100 ml 20% ILE and 0.25 ml/kg/min for an hour Co-interventions: IV fluids, calcium, atropine, adrenaline, HIET	Mortality: No Morbidity: Brady-asystolic arrest Haemodynamic improvement: Yes Secondary outcomes: Discharged day 11 Adverse effects: NR

(continued)

Continued.

Source	Subject and patient characteristics	Type of BB poisoning	Intervention and co-interventions	Results
Stellpflug et al. [120]	Subject: 30-year-old female Comorbidities: HOCM, implantable ICD Co-ingestions: Diltiazem, Amiodarone- unknown amount Status at arrival: Confused, hypotensive	Metoprolol – unknown amount	Intervention: 100 ml 20% ILE and 1.5 l over an hour Co-interventions: IV fluids, calcium gluconate, HIET	Mortality: No Morbidity: Renal impairment Haemodynamic improvement: Yes Secondary outcomes: Discharged to Psych ward day 5 Adverse effects: NR
Thompson et al. [122]	Subject: 7-month-old female Comorbidities: IUGR, scalp haemangioma, concurrent otitis media Co-ingestions: Nil Status at arrival: Decreased consciousness, lethargy, decreased muscle tone, HR 62	Propranolol 6 mg/kg × 3 propranolol instead of 0.6 mg/kg (unintentional 10 × error)	Intervention: 0.25 ml/kg 20% ILE and 0.0125 ml/kg/min for 2 h + another bolus 0.2 ml/kg Co-interventions: IV fluids, HIET	Mortality: No Morbidity: Nil Haemodynamic improvement: Yes Secondary outcomes: Discharged day 2 Adverse effects: NR
Troendle et al. (abstract) [121]	Subject: 55-year-old male Comorbidities: NR Co-ingestions: Amlodipine- unknown amount Status at arrival: Altered conscious state, hypotension, HR 57	Labetolol- unknown amount	Intervention: 0.167 mg/kg/min 20% ILE and 0.25 mg/kg/min infusion for 3 h Another 0.167 mg/kg/min 20% ILE over 90 min. Further ILE given but dosing reported inconsistent in case history Co-interventions: IV fluids, glucagon, calcium gluconate, calcium chloride, adrenaline, HIET, vasopressin, dopamine, adrenaline, phenylephrine	Mortality: No Morbidity: Acute renal impairment Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Walter et al. [123]	Subject: 51-year-old female Comorbidities: Hypertension Co-ingestions: Amlodipine 280 mg Status at arrival: Alert, hypotension, HR 50	Bisoprolol 140 mg	Intervention: 1.5 ml/kg 20% ILE followed by 15 ml/kg/h + second dose 10 h later Co-interventions: Glucagon, noradrenaline, calcium chloride, HIET	Mortality: No Morbidity: NR Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Lignocaine Bourg et al. [88]	Subject: 28-year-old female	Propranolol	Intervention: Single episode of VT treated with lidocaine 100 mg	Mortality: No Morbidity: NR Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Assimes and Malcolm [126]	Subject: 70-year-old male	Sotalol 3–4 g	Intervention: Torsades des Pointes (TdP), treated with 100 mg lidocaine for first episode and 50 mg for second episode	Mortality: No Morbidity: NR Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Belton et al. [127]	Subject: 57-year-old female	Sotalol 2.5 g	Intervention: Number of bouts of ventricular tachycardia after admission which were associated with convulsions. These were successfully treated with lignocaine	Mortality: No Morbidity: NR Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Edvardsson et al. [128]	Subject: 25-year-old female	Sotalol 15 g	Intervention: Lignocaine given for unconscious VT with no effect. Co-interventions: Defibrillation	Mortality: No Morbidity: NR Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Other treatments Wallin and Hulting [137]	Subject: 34-year-old male Status at arrival: Femoral pulse “barely palpable” first Hr, no BP readings given. Patient “stabilised” after that, HR 70–80, but no other CVS indices reported	Metoprolol 50 g	Intervention: Prenalterol bolus 10 mg, repeated 33 mg boluses during first hour (number not specified), then 120 mg over 14 h Co-interventions: Glucagon, atropine	Mortality: No Morbidity: NR Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Agura et al. [136]	Subject: 40-year-old female	Propranolol “Massive”	Intervention: Isoproterenol 160–200 µg/min Co-interventions: Glucagon 10 mg + infusion,	Mortality: No Morbidity: NR Haemodynamic improvement: Yes

(continued)

Continued.

Source	Subject and patient characteristics	Type of BB poisoning	Intervention and co-interventions	Results
Freestone et al. [135]	Subject: 53-year-old female	Atenolol – unknown amount	adrenaline, noradrenaline, dopamine, sodium bicarbonate infusions Intervention: Four 5 mg intravenous doses of prenalterol, over 30 min increased heart rate to 65 beats/min and blood pressure to 100/55 mmHg. Twelve hours after admission, given a further three × 5 mg of prenalterol intravenously followed by an infusion of 5 mg hourly in 5% (w/v) dextrose solution for 12 h for hypotension with maintenance of blood pressure. Co-interventions: Atropine, glucagon 10 mg boluses Intervention: Stabilized with atropine and adrenaline, glucagon, calcium and then the patient received a bolus injection of 0.5 mg of enoximone per kilogram of body weight, which resulted in an immediate increase in the cardiac output to 2.0 l per minute. The patient then received a continuous infusion of 15 mg of enoximone per kilogram per minute. Two hours later, the cardiac output was 3.6 l per minute, the stroke volume was 48 ml, and the systemic vascular resistance was 2058 dyn _{sec} cm ₅	Secondary outcomes: NR Adverse effects: NR Mortality: No Morbidity: NR Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Hoeper and Boeker [132]	Subject: 55-year-old female Status at arrival: Hypotension, BP 50 mmHg systolic	Metoprolol 10 g	Intervention: 260 mg isoproterenol over 65 h. 170 mg of this in first 12 h Co-interventions: Glucagon, dopamine	Mortality: No Morbidity: NR Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Hurwitz et al. [134]	Subject: 21-year-old female Co-ingestions: Trimipramine 625 mg	Acebutolol 9.6 g, Labetolol 7.2 g	Intervention: Amrinone infusion despite recovery of blood pressure Co-interventions: Infusions of glucagon, phenylephrine, dopamine Intervention: Enoximone bolus 1 mg/kg followed by infusion (0.5 µg/kg/min) 5 h post ingestion Co-interventions: Calcium boluses, dopamine, adrenaline infusions. Progressive improvement over next 6 h	Mortality: No Morbidity: NR Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Kollef [133]	Subject: 37-year-old female	Labetolol 800 mg	Intervention: 100-mg i.v. bolus of enoximone was administered and 3 min later recovery of spontaneous circulation occurred. Co-interventions: Amiodarone, 6 mg of epinephrine Intervention: 100-mg bolus of enoximone was given intravenously, followed by an infusion at a rate of 5 mg/kg/min. Co-interventions: ACLS, adrenaline, atropine, sodium bicarbonate, IV fluid	Mortality: No Morbidity: NR Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR Mortality: No Morbidity: NR Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Sandroni et al. [130]	Subject: 57-year-old male Co-ingestions: Verapamil 1600 mg	Atenolol 2.8 g		
Sandroni et al. [129]	Two cases: Subject: 38-year-old female. Subject: 50-year-old male	Propranolol 1600 mg, HCL acid, oxazepam		

(continued)

Continued.

Source	Subject and patient characteristics	Type of BB poisoning	Intervention and co-interventions	Results
Shanker et al. [131]	Subject: 24-year-old female Co-ingestions: Paroxetine 30 mg, Diazepam 5 mg × 28 Status at arrival: Cardiac arrest	Propranolol LA 80 mg × 92	Intervention: 300 mL 8.4% Na bicarbonate bolus Co-Interventions: ACLS, isoprenaline infusion, adrenaline (epinephrine) infusion, glucagon infusion, intravenous fluids	Mortality: No Morbidity: NR Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Dialysis Anthony et al. [141]	Subject: 49-year-old female Co-ingestions: Diltiazem 1200 mg	Metoprolol 500 mg	Intervention: Charcoal haemoperfusion 85 min + 120 min sequentially	Mortality: No Morbidity: NR Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Garg et al. [77]	Subject: 32-year-old male Co-ingestions: Digoxin 25 mg, insulin	Propranolol 2 g	Intervention: 15 h of charcoal haemoperfusion	Mortality: No Morbidity: NR Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Heise et al. [86]	Subject: 44-year-old female Co-ingestions: Chlorthalidone 2.25 g, lisinopril-unknown amount	Atenolol 4.5 g	Intervention: CVVHDF 16 h after ingestion for anuric renal failure and inability to eliminate atenolol Blood flow rate 12 l/h, a pre-pump dilution rate 0.75 l/h, and a dialysate flow rate 2 l/h. A pre-cartridge atenolol concentration of 12,750 µg/l (corrected for dilution) was collected with a dialysate concentration of 10,000 µg/l. It took 83 min to collect a 5 l bag of dialysate. This is equal to the removal of 50 mg of atenolol each 83 min during the first day Co-intervention Adrenaline, noradrenaline, ephedrine, vasopressin, glucagon, calcium, cardiac pacemaker, LV assist device, ECMO, sodium bicarbonate	Mortality: No Morbidity: NR Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Huang et al. [139]	Subject: 57-year-old male	Atenolol 2.8 g	Intervention: Haemodialysis was initiated approximately 11 h after the patient's ingestion. A polysulphone high efficiency high-flux dialyzer. Creatinine 250. Blood and dialysate flow rates were 400 and 500 ml/min. The treatment was 5 h. Second session of dialysis 10 h after first. Co-intervention IV fluid. Glucagon (14 mg), atropine (0.5 mg), dopamine (10 mg/kg/h), calcium gluconate (1 g), phenylephrine (800 mg), norepinephrine (15 mg/min), glucose (50 g), and insulin (10 units). Norepinephrine (14.9 mg/min), epinephrine (10.1 mg/min), vasopressin (2.4 units/h), and milrinone (0.25 mg/kg/min).	Mortality: No Morbidity: NR Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Pfaender et al. [138]	Subject: 45-year-old female Co-ingestions: Nifedipine – unknown amount	Atenolol – unknown amount	Intervention: CVVHDF was then started (Prisma CM HOSPAL system) with an ultrafiltration rate of 1500 ml per hour, which corresponded to a creatinine clearance of 25 ml/min. Atenolol upon admission of the patient to hospital was 69.6 µg/ml, which is 100-fold higher than the normal therapeutic	Mortality: No Morbidity: NR Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR

(continued)

Continued.

Source	Subject and patient characteristics	Type of BB poisoning	Intervention and co-interventions	Results
Rodríguez et al. [118]	Subject: 26-year-old male Comorbidities: Hypertension, renal transplant Co-ingestions: Amlodipine, lisinopril	Metoprolol	level Co-interventions: IV fluid, glucagon, calcium, atropine, dopamine, adrenaline, HIET Intervention: CVVHF unable to be used as ILE blocked filters Co-interventions: HIET, glucagon, calcium, dopamine, adrenaline, noradrenaline, vasopressin, ILE Intervention: 4 h of haemodialysis Co-interventions: Atropine, glucagon, naloxone, flumazenil, dopamine, isoprotenerol, normal saline, adrenaline, transvenous pacing, ECMO	Mortality: Yes Morbidity: NR Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR Mortality: No Morbidity: NR Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Rooney et al. [164]	Subject: 28-year-old female	Acebutolol		
Extracorporeal membrane oxygenation (ECMO) Bilbault et al. [143]	Subject: 38-year-old female Comorbidities: Nil Co-ingestions: Nil State at arrival: Unstable	Betaxolol 5.32 g	Intervention: ECMO Co-interventions: IV fluid, atropine, dobutamine, adrenaline, noradrenaline, glucagon	Mortality: No Morbidity: Left ventricular dysfunction Haemodynamic improvement: Yes Functional status at discharge: Femoral nerve paralysis Secondary outcomes: NR Adverse effects: Ischaemic leg Mortality: No Morbidity: No Haemodynamic improvement: yes Secondary outcomes: NR Adverse effects: NR
Chenoweth et al. [144]	Subject: 25-year-old female Comorbidities: Nil Co-ingestions: Diltiazem 30 g State at arrival: Unstable	Metoprolol – unknown amount	Intervention: VA ECMO Co-interventions: HIET, ILE, Glucagon, Calcium Chloride, pacing	Mortality: No Morbidity: No Haemodynamic improvement: yes Secondary outcomes: NR Adverse effects: NR
De Rita et al. [145]	Subject: 33-day-old female Comorbidities: Failure to thrive Co-ingestions: Nil State at arrival: Unstable	Propranolol 2.5 mg	Intervention: ECMO Co-interventions: Atropine, adrenaline, dopamine	Mortality: No Morbidity: Hyperkalaemia, metabolic acidosis Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Escajeda et al. [112]	Subject: 45-year-old male Comorbidities: Nil Co-ingestions: Nil State at arrival: Unstable	Metoprolol 10 g	Intervention: ECMO Co-interventions: Adrenaline, vasopressin, HIET, dobutamine	Mortality: No Morbidity: arrest Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Kacirova et al. [80]	Subject: 14-year-old female Comorbidities: Nil Co-ingestions: Propafenone 3 g State at arrival: Unstable	Metoprolol 1 g	Intervention: VA ECMO Co-interventions: Glucagon, adrenaline, isoprenaline, activated charcoal	Mortality: Yes Morbidity: PEA arrest Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: Ischaemic leg
Kolcz et al. [80]	Subject: 15-year-old female Comorbidities: Nil Co-ingestions: Verapamil 960 mg State at arrival: Unstable	Propranolol 550 mg	Intervention: ECMO Co-interventions: Gastric lavage, activated charcoal, dopamine, atropine, adrenaline, pacing, dobutamine, milrinone	Mortality: No Morbidity: Ventricular fibrillation Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR
Maskell et al. [146]	Subject: 17-year-old male Comorbidities: Nil Co-ingestions: Amlodipine-unknown amount State at arrival: Unstable	Metoprolol-unknown amount	Intervention: VA ECMO Co-interventions: Adrenaline, calcium, sodium bicarbonate, ILE, noradrenaline	Mortality: No Morbidity: cardiac arrest Haemodynamic improvement: yes Secondary outcomes: NR Adverse effects: NR
Nordmark Grass et al. [166]	Subject: 28-year-old female Comorbidities: Nil Co-ingestions: Amlodipine 1 g State at arrival: Unstable	Metoprolol 10 g	Intervention: ECMO Co-interventions: Noradrenaline, HIET, dobutamine	Mortality: No Morbidity: NR Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: NR

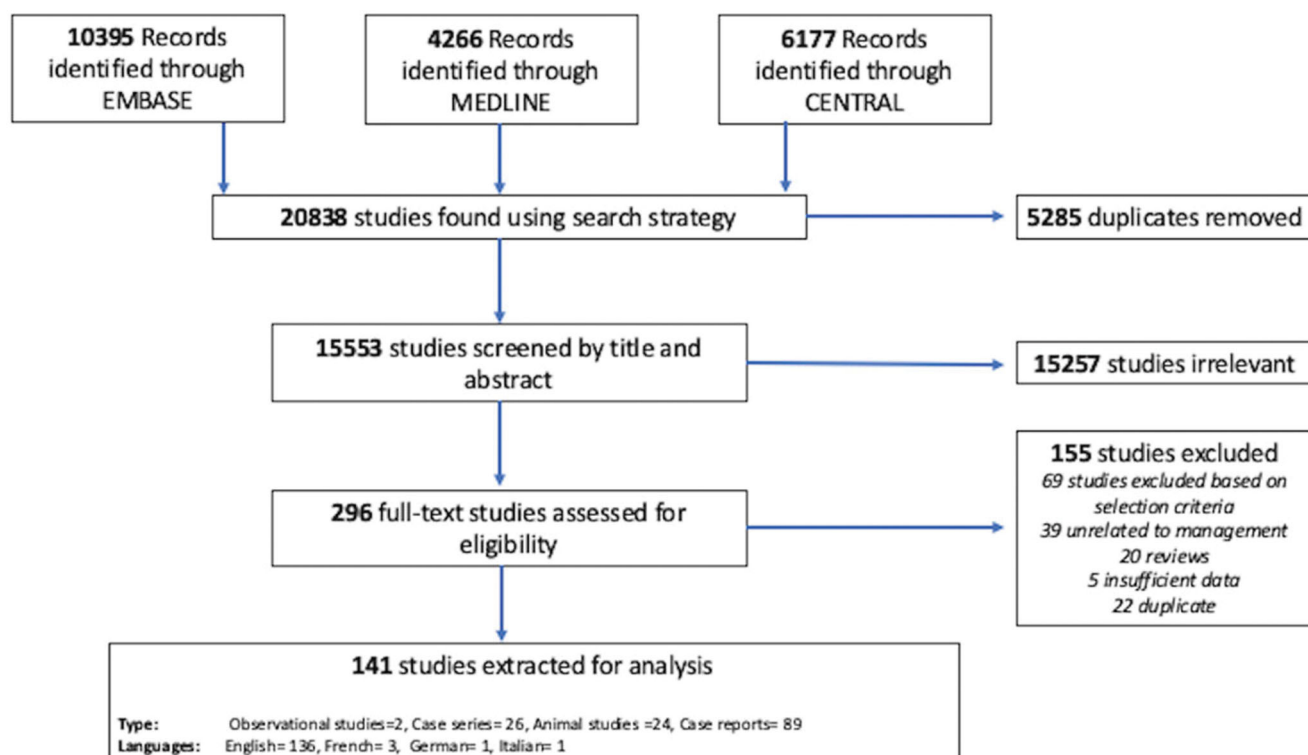
(continued)

Continued.

Source	Subject and patient characteristics	Type of BB poisoning	Intervention and co-interventions	Results
Rona et al. [147]	Subject: 36-year-old male Comorbidities: Nil Co-ingestions: Nifedipine State at arrival: Unstable	Atenolol 10 g	Intervention: ECMO Co-interventions: Dobutamine, adrenaline, calcium, glucagon, whole bowel irrigation, CVVHDF	Mortality: No Morbidity: metabolic acidosis, renal failure Haemodynamic improvement: yes Functional status at discharge: Reduced motor skills and impaired coordination Secondary outcomes: NR Adverse effects: NR
Rooney et al. [164]	Subject: 27-year-old female Comorbidities: Nil Co-ingestions: Nil State at arrival: Unstable	Acebutolol	Intervention: ECMO Co-interventions: Dopamine, dobutamine, glucagon, adrenaline, noradrenaline	Mortality: No Morbidity: Acute cardiomyopathy Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: deep vein thrombosis, ischemic hepatitis, acute tubular necrosis
Rygnestad et al. [165]	Subject: 29-year-old female Comorbidities: Nil Co-ingestions: 3.6 g Verapamil 3.6 g State at arrival: Unstable	Sotalol 4.8 g	Intervention: ECMO Co-interventions: HIET, CVVHDF	Mortality: No Morbidity: As below Haemodynamic improvement: Yes Secondary outcomes: NR Adverse effects: transient nerve paralysis, intestinal bleeding, compartment syndrome, rhabdomyolysis, renal failure
Transcutaneous pacing Belton et al. [127]	Subject: 57-year-old female Comorbidities: NR Co-ingestions: NR State at arrival: Unstable	Sotalol 2.5 g	Intervention: Temporary bipolar transcutaneous pacemaker Co-interventions: Lignocaine	Mortality: No Morbidity: seizure Haemodynamic improvement: yes Secondary outcomes: NR Adverse effects: NR
Intra-aortic balloon pump Lane et al. [148]	Subject: 17-year-old female Comorbidities: NR Co-ingestions: NR State at arrival: Unstable	Propranolol 8 g	Intervention: Intra-aortic balloon pump Co-interventions: Dopamine, dobutamine, atropine, glucagon, adrenaline, isoproterenol, bretyllium	Mortality: No Morbidity: Pulseless electrical activity Haemodynamic improvement: yes Secondary outcomes: NR Adverse effects: NR
Janion et al. [149]	Subject: 17-year-old female Comorbidities: T2 DM Co-ingestions: verapamil, amlodipine State at arrival: Unstable	Metoprolol – unknown amount	Intervention: Intra-aortic balloon pump Co-interventions: Adrenaline, noradrenaline, dobutamine	Mortality: No Morbidity: anuria Haemodynamic improvement: yes Secondary outcomes: NR Adverse effects: NR
Intra-cardiac device Heise et al. [86]	Subject: 44-year-old female Comorbidities: T2 DM Co-ingestions: chlorthalidone, lisinopril State at arrival: Unstable	Atenolol 4.5 g	Intervention: Left ventricular assist device Co-interventions: Phenylephrine, adrenaline, vasopressin, noradrenaline, esophagogastroduodenoscopy and removal stomach contents, CVVHDF,	Mortality: None Morbidity: acute respiratory distress Haemodynamic improvement: yes Secondary outcomes: NR Adverse effects: NR

ECMO: extracorporeal membranous oxygenation; CVVHDF: continuous venovenous haemodiafiltration; SDAC: single-dose activated charcoal; CPR: cardiopulmonary resuscitation; ALS: advanced life support; ILE: intravenous lipid emulsion; NR: not reported; U: units

Appendix 3. Flow diagram of articles found and included in beta-blocker systematic review



Appendix 4. Checklist adapted from guidance for the description of animal research in scientific publications [18]

Answer 'Yes' or 'No' for each item

Regarding the research animal, does the study...

1. Use good laboratory practices?
2. Specifically designed to evaluate treatment of beta-blocker (BB) poisoning?
3. Use unanaesthetised, unrestrained animals on semipurified diet?
4. Include adequate controls?
5. Use pigs, dogs or sheep for BB poisoning?

Regarding methodology, does the study...

6. Test multiple doses of test substance?
7. Estimate blood or other tissue level to ensure absorption and adequate exposure to active components to increase the likelihood the response will occur in humans?
8. Conduct clinical chemistry, blood and urine analysis?
9. Use more than a single species (and at least one non-rodent)?
10. Administer the BB orally?
11. Conduct animal necropsy and histopathology?
12. Use concealed, random allocation?
13. Blinded those assessing outcomes?
14. Use a sample size calculation or a power calculation?
15. Demonstrate statistical significance between groups?
16. Use appropriate statistical methods?

Appendix 5. Risks of bias across studies for each intervention. Extracorporeal membrane oxygenation (ECMO)

Treatment	Studies	Observational studies	Case series	Animal studies		
		STROBE (STrengthening the Reporting of OBservational studies in Epidemiology) and Thomas Tool	Institute of Health Economics (IHE) tool for quality of case series and quality of reporting (/20)	ARRIVE checklist (/20)	NRCNA checklist (/16)	Overall quality of evidence using grade criteria
Intravenous lipid emulsion	18 [20,21,23,24,27,28,40,43,45,46,48,49,52,59,60,66]	Smolinske et al. [66] STROBE: 19/22 Thomas tool: <i>Moderate:</i> Selection bias, study design, confounders <i>Weak:</i> Intervention integrity, data collection methods, blinding, withdrawals	6 ± 3 ^{a,b,c}	12 ± 2	4 ± 2	Very low
Insulin	14 [12,13,19,25,27,38,41–43,55,57,61,63,64]	–	10 ± 4 ^{a,b,c}	17 ± 2	9 ± 2	Low
Vasopressors	7 [12,13,32–32,44,57]	–	7 ± 2 ^{a,b,c}	12 ± 3	6 ± 1	Low
Glucagon	7 [29,30,32,36,53,55,65]	–	5 ± 1 ^{a,b,c}	11 ± 2	5 ± 2	Very low
ECMO	5 [31,34,35,39,62]	Masson et al. [31] STROBE: 17/22 Thomas tool: <i>Strong:</i> Selection bias <i>Moderate:</i> Cohort methods, blinding, withdrawal, confounding <i>Weak:</i> Intervention integrity	12 ± 3 ^{a,b,c}	–	–	Low
Calcium	3 [26,37,58]	–	–	17	8 ± 1	Very low
Levosimendan	2 [50,51]	–	–	17	6 ± 1	Very low
Milrinone	2 [29,53]	–	–	13	6 ± 1	Very low
Atropine	2 [22,33]	–	–	12 ± 1	4 ± 1	Very low
Pacing	1 [54]	–	11	–	–	Low
Amrinone	1 [30]	–	–	16	5	Very low
Fructose Diphosphate	1 [47]	–	–	9	5	Very low
Prenalterol	1 [56]	–	–	9	4	Very low
Isoprotenerol	1 [36]	–	–	5	3	Very low

Data represent the mean score and standard deviation obtained per intervention for each study (excluding case reports).

^aConfounding bias.

^bObserver bias.

^cSelection bias.